

Chapter 18 :

Voltage Regulators

18.1 Introduction

All electronic devices and circuits are designated to run at predetermined power ratings, i.e. voltage and current. While current consumption is dynamic and depends on the resistance of the devices or circuits, the voltage supply is fixed and ideally constant for the proper functioning of the devices and circuits. A voltage regulator is a circuit that creates and maintains a fixed output voltage, irrespective of changes to the input voltage or load conditions. The voltage regulator is an essential unit of a regulated power supply as shown in Fig.18-1.

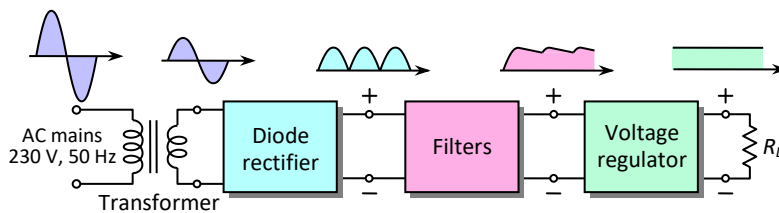


Fig.18-1: Block diagram of a regulated power supply

Voltage regulators (VRs) keep the voltages from a power supply within a range that is compatible with the other electrical components or circuits. While voltage regulators are most commonly used for DC-to-DC power conversion, some can perform AC-to-AC or AC-to-DC power conversion as well. Simple Zener diode voltage regulator circuits have been discussed in Chapter 2. In this chapter the voltage regulator will be discussed elaborately.

18.2 Power Supply Characteristics

The quality of a power supply depends on its load regulation, line regulation, and output resistance. In this section, we will look at these characteristics because they are often used on data sheets to specify power supplies.

Fig.18-2 shows an unregulated power supply that consists of a step-down transformer, a bridge rectifier and a capacitor filter. A problem of this power supply is that the change in load resistance will change the load (output) current and voltage. If we reduce the load resistance, we get more ripple and additional voltage drop across the transformer windings and diodes. Because of this, an

increase in load current always decreases the load voltage. In some datasheets, this property is represented by a parameter called **load effect**. Load effect is defined as the change in output voltage (ΔV_{Out}) for the change in load current from zero to rated maximum current ($I_{L(max)}$). Mathematically,

$$\text{Load effect} = \Delta V_{Out} \text{ for } \Delta I_L = I_{L(max)} \quad (18-1)$$

Here, the input line voltage has a nominal value of 220 V. The actual voltage coming out of a power outlet may vary from 200 V to 240 V RMS, depending on the time of day, the locality, and other factors. Since the secondary voltage is directly proportional to the line voltage, the load voltage in

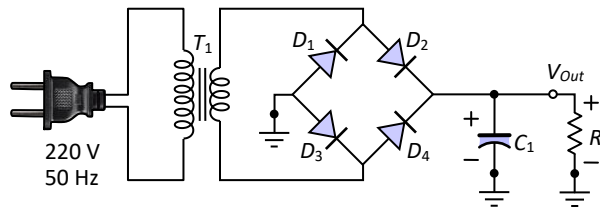


Fig.18-2: Unregulated power supply with capacitor filter

Fig.18-2 will change when line voltage changes. This property is represented by another parameter called **Source effect**. Source effect is defined as the change in output voltage (ΔV_{Out}) for 10% change in input voltage (source voltage). Mathematically,

$$\text{Line effect} = \Delta V_{Out} \text{ for } \Delta V_{In} = 0.1 V_{In} \quad (18-2)$$

For this reason, a voltage regulator is used in a power supply to keep the output voltage as constant as possible though the line voltage and load current may change.

Two basic functions of voltage regulation are line regulation and load regulation. The purpose of line regulation is to maintain a nearly constant output voltage when the input voltage varies. The purpose of load regulation is to maintain a nearly constant output voltage when the load varies.

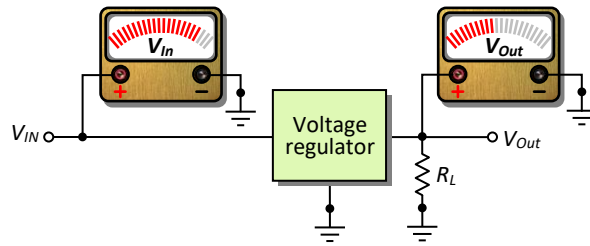
Line Regulation

When the AC input voltage (from grid line) of a power supply changes, the regulator maintains a nearly constant output voltage, as illustrated in Fig.18-1. The ability of a regulator to maintain a constant voltage irrespective of the input change is measured by a term called **line regulation**. Line regulation can be defined as the percentage change in the output voltage for a given change in the input voltage (generally 10%) with all other factors held constant. When taken over a range of input voltage values, line regulation is expressed as a percentage by the following formula:

$$\text{Line regulation} = \left(\frac{\Delta V_{Out}}{\Delta V_{In}} \right) \times 100\% \quad (18-3)$$

Line regulation can also be expressed in units of %/V. For example, a line regulation of 0.05%/V means that the output voltage changes 0.05% when the input voltage increases or decreases by one volt. Line regulation can be calculated using the following formula (Δ means a change):

$$\text{Line regulation} = \left(\frac{\Delta V_{Out}/V_{Out}}{\Delta V_{In}} \right) \times 100\% \quad (18-4)$$



Two voltmeters show the value of input and output voltage

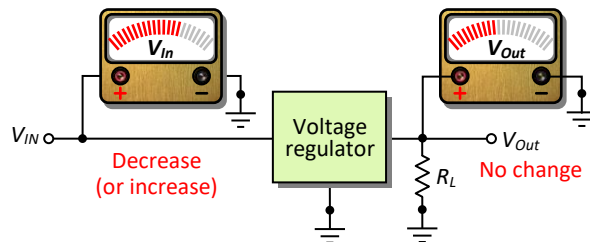


Fig.18-3: Line regulation. A change in input (line) voltage does not significantly affect the output voltage of a regulator (within certain limits)

Example 18-1

Due to the decrease in the AC input voltage of a certain power supply, the DC input to the voltage regulator decreases by 5 V, and the output of the regulator decreases by 0.25 V. The nominal output is 12 V. Determine the line regulation in %/V.

Solution:

Using Equ.(18-4),
$$\text{Line regulation} = \left(\frac{\Delta V_{out}/V_{out}}{\Delta V_{in}} \right) \times 100\%$$

or,
$$\text{Line regulation} = \left(\frac{0.25 \text{ V}/12 \text{ V}}{5 \text{ V}} \right) \times 100\% \approx 0.42\%/V \text{ [Ans.]}$$

Comments: The lower the value of line regulation the better the regulator quality.

Load Regulation

When the amount of current through a load changes due to variations in load resistance, the voltage regulator must maintain a nearly constant output voltage across the load, as illustrated in Fig.18-2.

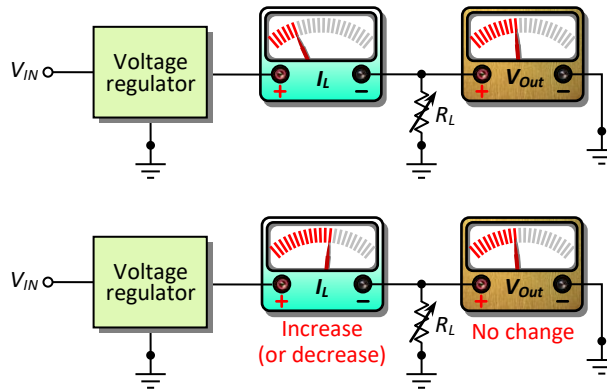


Fig.18-4: Load regulation. A change in load current has practically no effect on the output voltage of a regulator (within certain limits)

Load regulation can be defined as the percentage change in output voltage for a given change in load current with all other factors held constant. One way to express load regulation is as a percentage change in output voltage from no-load (NL) to full-load (FL) voltage. Mathematically,

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}} \right) \times 100\% \quad (18-5)$$

Alternately, the load regulation can be expressed as a percentage change in output voltage for each mA change in load current. For example, a load regulation of 0.01%/mA means that the output voltage changes (increase or decrease) by 0.01% when the load current increases or decreases by 1 mA.

Example 18-2

A certain voltage regulator has 12 V output when there is no load ($I_L = 0$). When the full-load current of 200 mA is drawn from regulator, the output voltage falls to 11.8 V. Calculate the voltage regulation as a percentage change from no-load to full-load and also as a percentage change for each mA change in load current.

Solution:

Here, the no-load voltage is $V_{NL} = 12$ V, and the full-load voltage is $V_{FL} = 11.8$ V. The voltage regulation (load regulation) can be calculated using Eq.(18-5) as,

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}} \right) \times 100\% = \left(\frac{12 \text{ V} - 11.8 \text{ V}}{11.8 \text{ V}} \right) \times 100\%$$

$$\therefore \text{Load regulation} \approx 1.7\% \text{ [Ans.]}$$

The load regulation can also be calculated as a percentage change per mA of output current as,

$$\text{Load regulation} = \frac{1.7\%}{200 \text{ mA}} \approx 0.0085\%/\text{mA} \text{ [Ans.]}$$

Comments: The smaller the load regulation, the better the power supply.

Sometimes power supply manufacturers specify the equivalent output resistance of a power supply (R_o) instead of its load regulation. Recall that an equivalent Thevenin circuit can be drawn for any two-terminal linear circuit. Fig.18-3 shows the equivalent Thevenin circuit for a power supply with a load resistor (R_L). The Thevenin voltage is the voltage from the supply with no load (V_{NL}), and the Thevenin resistance is the specified output resistance, R_o . Ideally, R_o should be zero, corresponding to 0% load regulation. However, in practical power supplies R_o has a small value. With the load resistor in place, the output voltage is found by applying the voltage-divider rule as,

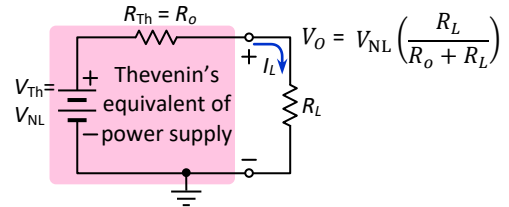


Fig.18-5: Thevenin equivalent circuit for a power supply with a load resistor

$$V_O = V_{NL} \left(\frac{R_L}{R_o + R_L} \right) \quad (18-6)$$

If we let R_{FL} equals the smallest-rated load resistance (for rated output current), then the full-load output voltage (V_{FL}) will be,

$$V_{FL} = V_{NL} \left(\frac{R_{FL}}{R_o + R_{FL}} \right)$$

or,

$$V_{NL} = V_{FL} \left(\frac{R_o + R_{FL}}{R_{FL}} \right)$$

Putting this value of V_{NL} into Equ.(18-3) we get,

$$\text{Load regulation} = \frac{V_{FL} \left(\frac{R_o + R_{FL}}{R_{FL}} \right) - V_{FL}}{V_{FL}} \times 100\%$$

$$\text{Load regulation} = \left(\frac{R_o + R_{FL}}{R_{FL}} - 1 \right) \times 100\%$$

$$\boxed{\text{Load regulation} = \left(\frac{R_o}{R_{FL}} \right) \times 100\%} \quad (18-7)$$

Thus using Equ.(18-5), we can easily find the percentage of load regulation when the output resistance, and the minimum load resistance of a power supply is specified.

18.3 Types of Voltage Regulator

There are two main types of voltage regulators: **linear voltage regulator** and **switching or switchmode voltage regulator**. Though both types regulate a system's voltage, but linear regulators operate with lower efficiency and switching regulators operate with higher efficiency. In

high-efficiency switching regulators, most of the input power is transferred to the output without dissipation.

18.3.1 Linear Regulators

A linear voltage regulator utilizes an active pass device (such as a BJT or MOSFET), which is controlled by a high-gain (BJT or op-amp) amplifier. To maintain a constant output voltage, the linear regulator adjusts the pass device resistance by comparing the internal voltage reference to the sampled output voltage, and then driving the error to zero. Linear regulators are step-down converters, so by definition the output voltage is always below the input voltage. However, these regulators offer a few advantages: they are generally easy to design, dependable, cost-efficient, and offer low noise as well as a low output voltage ripple.

Linear regulators, such as the MP2018, only require an input and output capacitor to operate (see Fig.18-xx). Their simplicity and reliability make them intuitive and simple devices for engineers, and are often highly cost-effective.

18.3.2 A Switching Regulator

A switching regulator circuit is generally more complicated to design than a linear regulator, and requires selecting external component values, tuning control loops for stability, and careful layout design. Switching regulators can be step-down converters, step-up converters, or a combination of the two, which makes them more versatile than a linear regulator. Advantages of switching regulators include that they are highly efficient, have better thermal performance, and can support higher current and wider V_i/V_o applications. They can achieve greater than 95% efficiency depending on the application requirements. Unlike linear regulators, a switching power supply system may require additional external components, such as inductors, capacitors, FETs, or feedback resistors. The HF920 is an example of a switching regulator that offers high reliability and efficient power regulation (see Fig.18-xx).

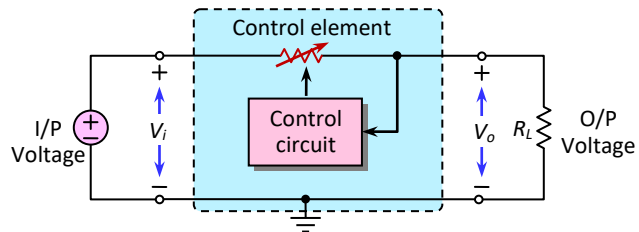


Fig.18-6: Block diagram of a linear voltage regulator

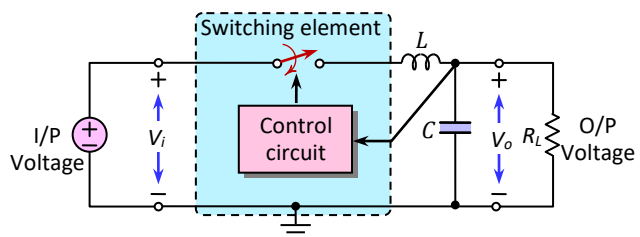


Fig.18-7: Block diagram of a switching type voltage regulator

18.4 Basic Linear Voltage Regulators

The fundamental classes of voltage regulators are linear regulators and switching regulators. Both of these are available in integrated circuit form. Two basic types of linear regulator are the *series regulator* and the *shunt regulator*.

18.5 Transistor Series Regulators

A block diagram of a series type of linear regulator is shown in Fig.18-8. The output sample circuit senses a change in the output voltage. The error detector compares the sample voltage with a reference voltage and causes the control element to compensate in order to maintain a constant output voltage.

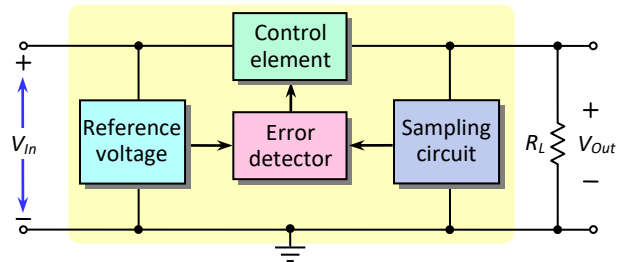


Fig.18-8: Block diagram of a series type voltage regulator

18.5.1 Basic Circuit

In Chapter 2, we have discussed a basic voltage regulator (shunt type) using Zener diode. In this type of regulator, the load current is limited by the maximum Zener diode current capacity.

The current capacity can be increased by using a series-pass transistor as shown in Fig.18-9. This circuit is a *series voltage regulator*. Transistor Q_1 is referred to a *series-pass transistor*.

The output voltage (V_{Out}) of this regulator is ($V_Z - V_{BE}$), and the maximum load current $I_{L(max)}$, can be the maximum emitter current of Q_1 . For a 2N3055 transistor (Specification is given in), I_L can be as much as 15 A.

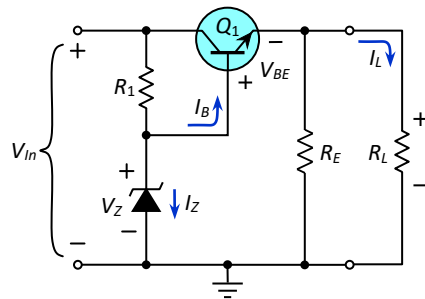


Fig.18-9: Block diagram of a series type voltage regulator

18.5.2 Regulator With Error Amplifier

A series regulator with an *error amplifier* is shown in Fig.18-10. The error amplifier improves the line regulation and load regulation. The amplifier also makes it possible to have an output voltage greater than the Zener diode voltage. Resistor R_2 and Zener diode D_1 produce the reference voltage (V_Z). Transistor Q_2 and R_1 constitute the error amplifier, that controls the series-pass transistor Q_1 . The output voltage is divided by R_3 , and R_4 resistors and compared to the reference voltage, V_Z . C_1 is a large value capacitor, usually 50 μF , used to suppress any oscillation at the output.

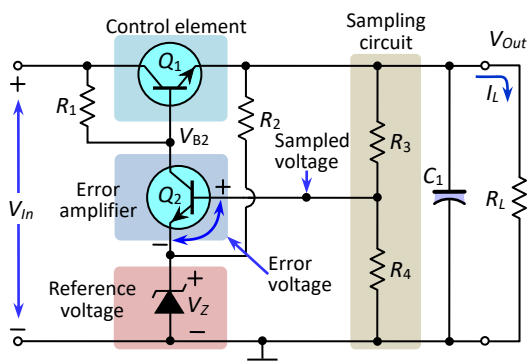


Fig.18-10: Block diagram of a series type voltage regulator

When the circuit output voltage changes, the change is amplified by transistor Q_2 and fed back to the base of Q_1 to correct the output voltage level. Suppose that the circuit is designed for $V_{Out} = 12\text{ V}$. The supply voltage (V_{In}) should be greater than V_{Out} , say $V_{In} = 18\text{ V}$. Let, $V_Z = 6\text{ V}$. The base voltage of Q_2 must be, $\therefore V_{B2} = V_Z + V_{BE2} = 6.7\text{ V}$. Resistors R_3 and R_4 will divide $V_{Out} = 12\text{ V}$ and produce $V_{B2} = 6.7\text{ V}$. The voltage at the base of Q_1 is, $V_{B1} = V_{Out} + V_{BE1} = 12.7\text{ V}$. Also, $V_{R1} = V_{In} - V_{B1} = 5.3\text{ V}$.

Now suppose (for any reason) the output voltage drops slightly. When V_{Out} drops, V_{B2} also drops. As the emitter voltage of Q_2 is held at V_Z , any decrease in V_{B2} , appears across the base-emitter of Q_2 . A reduction in V_{BE2} causes I_{C2} to be reduced. When I_{C2} falls, the voltage drop across R_1 also drops. So the voltage at the base of Q_1 rises ($V_{B1} = V_{In} - V_{R1}$) causing the output voltage to increase. Thus, a decrease in V_{Out} produces a feedback effect which causes V_{Out} to increase back to its normal level. In the same process, a rise in V_{Out} above its normal level produces a feedback effect which pushes V_{Out} down again to its normal level.

Practically, with the change in input voltage (ΔV_{In}), a small change in output voltage (ΔV_{Out}) will take place. This change will depend on the gain of the error amplifier as,

$$\Delta V_{Out} = \frac{\Delta V_{In}}{A_v} \quad (18-8)$$

Example 18-3

The input voltage for the regulator in Fig.18-9 has $V_{In} = 20\text{ V}$ on no load, and $V_{In} = 19\text{ V}$ on full load ($I_{L(max)} = 50\text{ mA}$). The nominal output voltage is $V_{Out} = 12\text{ V}$, and the regulator has an error amplifier with a gain of $A_v = 100$. Calculate the load effect, and load regulation, line effect, and line regulation, for this power supply. **Also determine the ripple rejection ratio in decibels.**

Solution:

Using Equ.(18-1), Load effect = ΔV_{Out} for $\Delta I_L = I_{L(max)}$

Here, $\Delta V_{In} [\text{for } \Delta I_L = I_{L(max)}] = 20\text{ V} - 19\text{ V} = 1\text{ V}$

For this input change, the change in output voltage can be calculated using Equ.(18-8).

Using Equ.(18-8),
$$\Delta V_{Out} = \frac{\Delta V_{In}}{A_v} = \frac{1 \text{ V}}{100} = 10 \text{ mV [Ans.]}$$

% load regulation can be calculated using equ.(18-5).

Using Equ.(18-5),
$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}} \right) \times 100\% = 8.33\% \text{ [Ans.]}$$

Using Equ.(18-2),
$$\text{Line effect} = \Delta V_{Out} \text{ for } \Delta V_{In} = 10\% \text{ of } V_{In}$$

Here,
$$\Delta V_{In} = 10\% \text{ of } 20 \text{ V} = 2 \text{ V}$$

$$\therefore \text{Line effect} = \Delta V_{Out} = \frac{10\% \text{ of } V_{In}}{A_v}$$

$$\therefore \text{Line effect} = \Delta V_{Out} = \frac{2 \text{ V}}{100} = 20 \text{ mV [Ans.]}$$

Using Equ.(18-4),
$$\text{Line regulation} = \left(\frac{\Delta V_{Out}/V_{Out}}{\Delta V_{in}} \right) \times 100\%$$

$$\therefore \text{Line regulation} = \left(\frac{20 \text{ mV}/12 \text{ V}}{2 \text{ V}} \right) \times 100\% = 0.08\%$$

Ripple rejection ratio can be calculated as

$$\text{Ripple rejection ratio} = 20 \log(1/A_v)$$

or,
$$\text{Ripple rejection ratio} = 20 \log(1/100) = -40 \text{ dB [Ans.]}$$

Comments: With the change in load current, the input voltage of the regulator can be change due to the internal resistance of the rectifier circuit.

18.5.3 Regulator Design

To design a series regulator circuit (as in Fig. 17-10), the Zener diode is selected to have V_Z less than the output voltage. A Zener diode voltage approximately equal to $0.75 \times V_{Out}$ is usually suitable. Appropriate current levels are chosen for each resistor, and the resistor values are calculated using Ohm's law. Transistor Q_1 is selected to pass the required load current and to survive the necessary power dissipation. A heat sink (see Section 8-8) is normally required for the series-pass transistor (Q_1) in a regulator that supplies large load currents. As discussed, a large capacitor (C_1 in Fig. 17-10) is usually connected across the output to reduce output voltage fluctuations.

The difference between the regulator input and output voltages is the collector-emitter voltage of the series-pass transistor (Q_1), and this

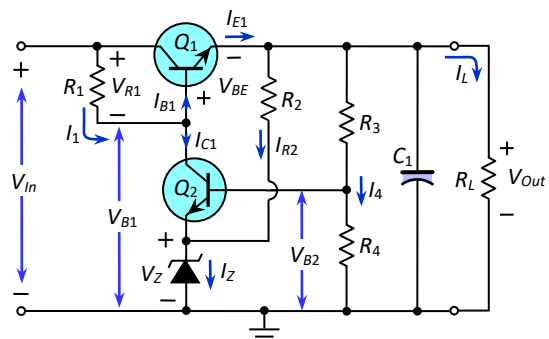


Fig.18-11: Block diagram of a series type voltage regulator

voltage must be large enough to keep the transistor operational. The minimum level of V_{CE1} (known as the dropout voltage) occurs at the lowest point in the ripple waveform of (rectified and filtered) raw DC input. If V_{CE1} is too small for correct operation at this point, a large-amplitude ripple waveform appears at the regulator output.

Example 18-4

Design a series voltage regulator as shown in Fig.18-10, to produce $V_{Out} = 12\text{ V}$ with maximum output load current $I_{L(max)} = 100\text{ mA}$. Assume the supply voltage is $V_{In} = 20\text{ V}$, minimum current gain of the series-pass transistor is $h_{FE1(min)} = 50$ and that of the error amplifier transistor is $h_{FE2(min)} = 120$.

Solution:

Let $V_Z = 0.5 V_{Out} = 0.5 \times 12\text{ V} = 6\text{ V}$. So, we can use 1N4735A that has $V_Z = 6.2\text{ V}$ at 40 mA [see Appendix] [H_A_atten].

Let us select the minimum current of the Zener diode is $I_{R2} = 5\text{ mA}$. Therefore,

$$R_2 = \frac{V_{R2}}{I_{R2}} = \frac{V_{Out} - V_Z}{I_{R2}} = \frac{12\text{ V} - 6\text{ V}}{5\text{ mA}} = 1.2\text{ k}$$

Neglecting I_4 , $I_{E1(max)} \approx I_{L(max)} + I_{R2} = 100\text{ mA} + 5\text{ mA} = 105\text{ mA}$.

Specifications of Q_1 :

$V_{CE1(max)} = V_{In} = 20\text{ V}$, $I_{C1(max)} \approx 105\text{ mA}$, and

$$P_{D1(max)} = (V_{In} - V_{Out})I_{E1(max)}$$

or,

$$P_{D1(max)} = (20\text{ V} - 12\text{ V}) \times 105\text{ mA} = 840\text{ mW}$$

The maximum base current of Q_{B1} will be,

$$I_{B1(max)} = \frac{I_{E1(max)}}{h_{FE1(min)}} = \frac{105\text{ mA}}{50} = 2.1\text{ mA}$$

The collector current of Q_2 should be much larger than $I_{B1(max)}$. Let $I_{C2} = 10\text{ mA}$. Therefore,

$$R_1 = \frac{V_{In} - V_{B1}}{I_{C2} + I_{B1}} = \frac{20\text{ V} - (12 + 0.7)\text{V}}{10\text{ mA} + 2.1\text{ mA}} \approx 603\ \Omega \text{ [use } 560\ \Omega \text{ plus } 43\ \Omega]$$

The maximum current of the Zener diode will be,

$$I_Z = I_{E2} + I_{R2} \approx I_{C2} + I_{R2} = 10\text{ mA} + 5\text{ mA} = 15\text{ mA}$$

Therefore, the power capacity of the Zener diode should be,

$$P_Z = V_Z I_Z = 6.2\text{ V} \times 15\text{ mA} = 93\text{ mW}$$

The base current of Q_2 will be,

$$I_{B2(max)} = \frac{I_{C2(max)}}{h_{FE2(min)}} = \frac{10\text{ mA}}{120} = 17.5\ \mu\text{A}$$

I_4 should be much larger than $I_{B2(max)}$. Let us select $I_4 = 1\text{ mA}$. Therefore,

$$R_3 = \frac{V_{Out} - V_{B2}}{I_4} = \frac{12 \text{ V} - (6.2 \text{ V} + 0.7 \text{ V})}{1 \text{ mA}} = 5.1 \text{ k}$$

$$R_4 = \frac{V_{B2}}{I_4} = \frac{6.2 \text{ V} + 0.7 \text{ V}}{1 \text{ mA}} = 6.9 \text{ k} \text{ [use standard } 6.8 \text{ k]}$$

The designed regulator is shown in Fig.18-11.

Comments: C_1 is a power supply decoupling capacitor and its value can be $10 \mu\text{F}$ to $100 \mu\text{F}$.

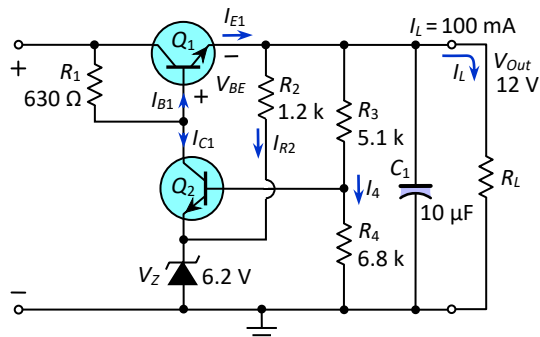


Fig.18-12: Voltage regulator designed in Example 18-4

18.6 Regulator with Variable Voltage

Regulator circuits such as the one designed in Example 17-2 are unlikely to have V_o exactly as specified. This is because of component tolerances, as well as, perhaps, not finding standard values close to the calculated values. Hence, some form of adjustment is required to enable the output voltage to be set to the desired level. In Fig. 17-5 the potentiometer (R_5) connected between resistors R_2 and R_4 provides output adjustment. The minimum output voltage level is produced when the potentiometer moving contact (wiper) is at the top of R_5 (as shown in Fig.18-14). The minimum voltage level will be given by,

$$V_{o(min)} = \frac{R_3 + R_4 + R_5}{R_4 + R_5} \times V_{B2} \quad (18-9)$$

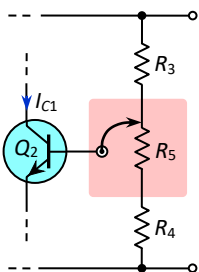


Fig.18-14: Position of potentiometer for minimum output voltage

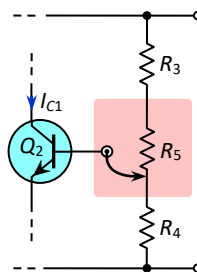


Fig.18-15: Position of potentiometer for maximum output voltage

Minimum V_o occurs when the moving contact is at the top of R_5 (as shown in Fig.18-15). The maximum output can be calculated as,

$$V_{O(max)} = \frac{R_3 + R_4 + R_5}{R_4} \times V_{B2} \quad (18-10)$$

Example 18-5

Design a voltage regulator to produce variable output voltage $V_{Out} = 12\text{ V} \pm 25\%$ with maximum output load current $I_{L(max)} = 100\text{ mA}$. Assume the supply voltage is $V_{In} = 20\text{ V}$, minimum current gain of the series-pass transistor is $h_{FE1} = 50$ and that of the error amplifier transistor is $h_{fe2} = 120$.

Solution:

The requirement of this design is exactly same to that of Exmaple 18-4. Only the difference is the variable output voltage. So, all components will remain same except the sampling unit. Here, we have to add a variable resistance, R_5 .

The maximum output voltage will be $V_{Out(max)} = 12\text{ V} + 12 \times 0.25\text{ V} = 15\text{ V}$, and the minimum output voltage will be, $V_{Out(min)} = 12\text{ V} - 12 \times 0.25\text{ V} = 9\text{ V}$

From Example 18-4 we get, $V_{B2} = V_Z + V_{BE2} = 6.2\text{ V} + 0.7\text{ V} = 6.9\text{ V}$. Therefore for $V_{Out} = 9\text{ V}$, moving the contact of R_5 to the top [Fig.18-15],

$$R_3 I_3 = V_{O(min)} - V_{B2}$$

As the base current of Q_2 is very small, we may assume that the current remains same in all resistors of the sampling branch. Let $I_{3(min)} = I_{4(min)} = 1\text{ mA}$ for the minimum output voltage.

$$\therefore R_3 = \frac{V_{O(min)} - V_{B2}}{I_{4(min)}} = \frac{9\text{ V} - 6.9\text{ V}}{1\text{ mA}} = 2.1\text{ k} \text{ [use standard } 2.0\text{ k]}$$

$$\text{Similarly, } R_5 + R_4 = \frac{V_{B2}}{I_{4(min)}} = \frac{6.9\text{ V}}{1\text{ mA}} = 6.9\text{ k}$$

For $V_O = 15\text{ V}$, the contact of R_5 has to be placed to the bottom [Fig.18-15]. For this voltage the value of $I_{3(max)} = I_{4(max)}$ will be,

$$I_{3(max)} = I_{4(max)} = \frac{V_{O(max)}}{R_3 + R_5 + R_4} = \frac{15\text{ V}}{2\text{ k} + 6.9\text{ k}} \approx 1.7\text{ mA}$$

$$R_4 I_{4(max)} = V_{O(max)} - (V_{R3} + V_{R5}) = V_{B2}$$

$$\text{or, } R_4 = \frac{V_{B2}}{I_{4(max)}} = \frac{6.9\text{ V}}{1.7\text{ mA}}$$

$$\therefore R_4 = 4.1\text{ k} \text{ [use standard } 4.3\text{ k]}$$

Therefore, $R_5 = (R_5 + R_4) - R_4 = 6.9\text{ k} - 4.3\text{ k} = 2.6\text{ k}$ [use standard 2.7 k potentiometer].

Comments: The designed voltage regulator is shown in Fig.18-15.

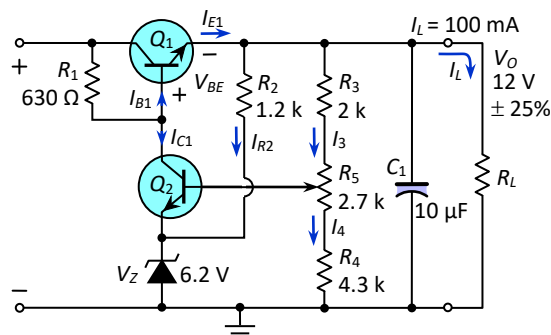


Fig.18-16: Voltage regulator designed in Example 18-5

18.7 Voltage Regulator with High Output Current

If we need high output current (1 A or above), the circuit of Fig.18-10 will not work properly. For high output current, the base current of series-pass transistor, Q_1 , will be very high, and hence the collector current of Q_2 has to be set to higher value. For this reason current through R_1 will also be high. As a result the value of R_1 will be very small. This small value of R_1 will result in low gain (A_v) of the error amplifier. According to Equ.(18-8), the lower value of A_v will deteriorate the load regulation and line regulation property of the regulator circuit. To overcome this problem, we can use a Darlington pair in place of the series-pass transistor (Q_1). The circuit arrangement is shown in Fig.18-16. Here, transistor Q_1 will be a high-power transistor, that will carry almost the entire output current and the transistor Q_3 will be a low-power normal transistor. As most of the current will flow through Q_3 , a suitable heatsink must be attached with it. As described in Chapter 5, the effective current gain of the Darlington pair will be,

$$h_{FE(D)} \approx h_{FE1} \times h_{FE3} \quad (18-11)$$

The base current of Q_3 will be,

$$I_{B3} \approx \frac{I_{L(max)} + I_{R2}}{h_{FE(D)}} = \frac{I_{L(max)} + I_{R2}}{h_{FE1} \times h_{FE3}} \quad (18-12)$$

Now, the value of I_{B3} will be very small and hence, a small value of I_{C2} will work properly. For this small value of I_{C3} , I_{R1} will also be small which results in a larger value of R_1 . This larger value of R_1 will give larger value of A_v and consequently, the regulator will work with better load and line regulation. The resistor R_6 in Fig.18-16, is used to provide a path for a suitable quiescent current for Q_3 , when the output collector current of Q_1 is very small.

Instead of using two transistors in Darlington configuration, we can also use **Darlington power transistors**. These transistors contain two transistors connected as Darlington pair, in a single package. For example, TIP120 is a Darlington power transistor that has $h_{FE(min)} = 1000$, and $I_{C(max)} = 5$ A. Remember that the base-emitter voltage of this transistor will be just double the single transistor, that is $V_{BE} = 1.4$ V. As a Darlington transistor has three terminals, there is no way to use R_6 resistor.

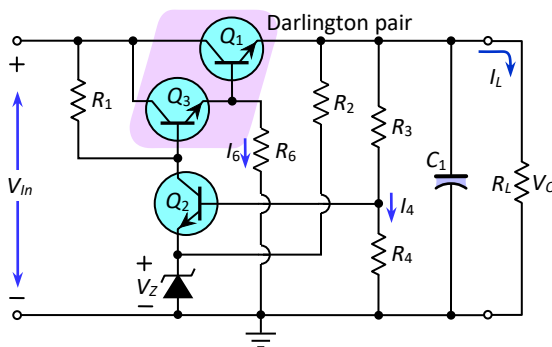


Fig.18-17: Block diagram of a series type voltage regulator

Example 18-6

Design a voltage regulator as shown in Fig.18-, to produce $V_{out} = 12$ V with maximum output load current $I_{L(max)} = 1$ A. Assume the supply voltage is $V_{in} = 20$ V, the minimum current gains of the transistors are $h_{FE1} = 50$, and $h_{FE2} = h_{FE3} = 120$.

Solution:

Neglecting the current in the sampling branch (I_4), the maximum emitter current of Q_1 will be,

$$I_{E1(max)} \approx I_{L(max)} + I_{R2} = 1000 \text{ mA} + 5 \text{ mA} = 1005 \text{ mA}$$

The specifications of Q_1 should be:

$V_{CE1(max)} = V_{In} = 20 \text{ V}$, $I_{C1(max)} \approx 1005 \text{ mA}$, and

$$P_{D1(max)} = (V_{In} - V_{Out})I_{E1(max)}$$

or,

$$P_{D1(max)} = (20 \text{ V} - 12 \text{ V}) \times 1005 \text{ mA} = 8.4 \text{ mW}$$

The maximum base current of Q_1 will be,

$$I_{B1(max)} = \frac{I_{E1(max)}}{h_{FE1(min)}} = \frac{1005 \text{ mA}}{50} \approx 20 \text{ mA}$$

The collector current of Q_3 should be a little larger than $I_{B1(max)}$. Let $I_{C3} = 22 \text{ mA}$ ($I_{B1} = 20 \text{ mA}$, and $I_6 = 2 \text{ mA}$). Therefore,

$$I_{B3(max)} = \frac{I_{C3(max)}}{h_{FE3(min)}} = \frac{22 \text{ mA}}{120} \approx 0.18 \text{ mA}$$

The collector current of Q_2 should be much larger than $I_{B3(max)}$. Let $I_{C2} = 2 \text{ mA}$. Therefore,

$$R_1 = \frac{V_{In} - V_{B3}}{I_{C2} + I_{B3}} = \frac{V_{In} - (V_O + V_{BE1} + V_{BE2})}{I_{C2} + I_{B3}}$$

or,

$$R_1 = \frac{20 \text{ V} - (12 + 0.7 + 0.7) \text{ V}}{2 \text{ mA} + 0.18 \text{ mA}} \approx 3 \text{ k}$$

Value of R_6 can be calculated as,

$$R_6 = \frac{V_O + V_{BE1}}{I_6} = \frac{12 \text{ V} + 0.7 \text{ V}}{2 \text{ mA}} \approx 6.4 \text{ k} [\text{use } 6.2 \text{ k}]$$

Rest of the design process will be same as Example 18-.

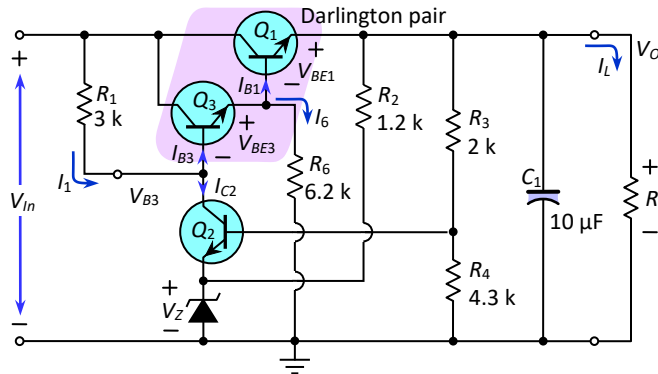


Fig.18-18: Voltage regulator designed in Example 18-6

The designed regulator is shown in Fig.18-17.

Comments: C_1 is a power supply decoupling capacitor and its value can be $10 \mu\text{F}$ to $100 \mu\text{F}$.

18.8 Protection Circuits for Voltage Regulators

Voltage regulators may be used with a fixed load in electronic systems, or in general purpose power supplies with variable loads (e.g. power supplies used in laboratories). The general purpose power supplies used in laboratories may be accidentally shorted or may be overloaded by the users. The overload currents or short-circuit may destroy the series-pass power transistor, or even may damage the loads. The series regulator described so far has no protection circuit. Therefore, if we accidentally short the load terminals, the load current will try to approach infinity (for high current circuit like in Fig.18- Darlington pair), which may destroy the series-pass transistor (Q_1). It may also destroy one or more diodes in the rectifier circuit of unregulated power supply that drives the series regulator or even burn out the transformer. To protect against an accidental short-circuit or overloading, series regulators usually include some form of **current limiting** circuits. The specially designed current limiting circuitry limits the current that can be drawn from the supply to a certain specific limit [$I_{L(max)}$], even if the output terminals are short-circuited. There are mainly two types of current limiting circuits:

18.8.1 Constant Current Limiting

Constant current limiting technique is the most common and simple form of current limiting used in regulated power supplies. In this technique, the output voltage of the power supply remains constant as the output current increases until the maximum current limit is reached. As the output current exceeds the maximum current limit, the voltage falls down rapidly with increasing load, but the load current is maintained at the same level (a little decrease with the increase of load current) as shown in Fig.18-19. The main disadvantage of this current limiting circuit is the power dissipation in the series-pass transistor in overload/short-circuit conditions. The amount of current through, and the voltage across this transistor becomes excessively high. This problem is little bit improved in fold-back current limiting system.

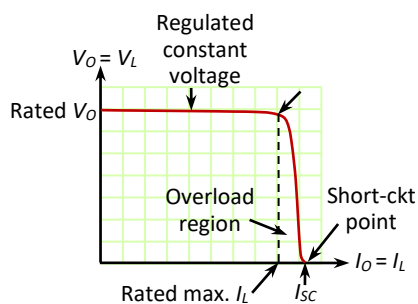


Fig.18-19: Graph of constant current limiting power supply

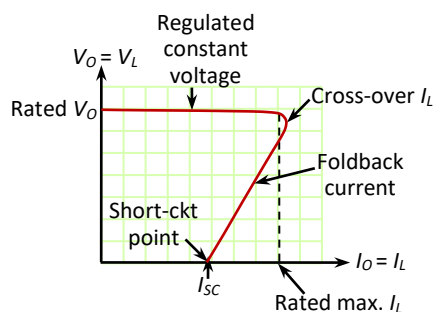


Fig.18-20: Graph of constant current limiting power supply

18.8.2 Fold-back Current Limiting

Though, a little bit complicated, fold-back current limiting is commonly used in voltage regulator ICs. Like the constant current limiting, here also, the output voltage supply remains constant until the maximum current limit is reached. However, beyond the maximum current limit, rather than

maintaining the same, the current starts to reduce as shown in Fig.18-20. Foldback current limiting in a voltage regulator reduces the power consumption, because as the overload increases, so the current folds back and the overall power consumption falls, reducing power dissipation in the series pass transistor.

18.8.3 Voltage Regulator with Constant Current Limiting Circuit

Fig.18-21 shows a voltage regulator with a popular current limiting circuit consisting of a transistor Q_4 and a resistor R_{SC} . Here, R_{SC} is called **current-sensing resistor**. The load current flows through R_{SC} and hence, with the increase in I_L the voltage drop across R_{SC} increases. The value of R_{SC} is so chosen that, for $I_L \leq I_{L(max)}$, the voltage drop across R_{SC} is not large enough to turn the transistor Q_4 ON. That is, for this current, V_{R4} will not be sufficient to forward bias the $B-E$ junction of Q_4 . Thus, Q_4 has not effect on the operation of the regulator circuit. So, the regulator circuit works normally as described before for the load current upto $I_{L(max)}$. However, when the load current exceeds the maximum limit [$I_L > I_{L(max)}$], the voltage drop across R_{SC} becomes sufficient to forward bias the $B-E$ junction of transistor Q_4 and turns this transistor ON. In this case, transistor Q_4 will draw a substantial collector current (I_{C4}) flowing through R_1 . As a consequence, current that would otherwise enter the base of Q_1 is diverted through Q_4 . In this way, the series-pass transistor is prevented from supplying additional load current. When high current flows through R_{SC} , the voltage drop across R_{SC} and Q_1 (V_{CE1}) will increase and the output voltage will also decrease.

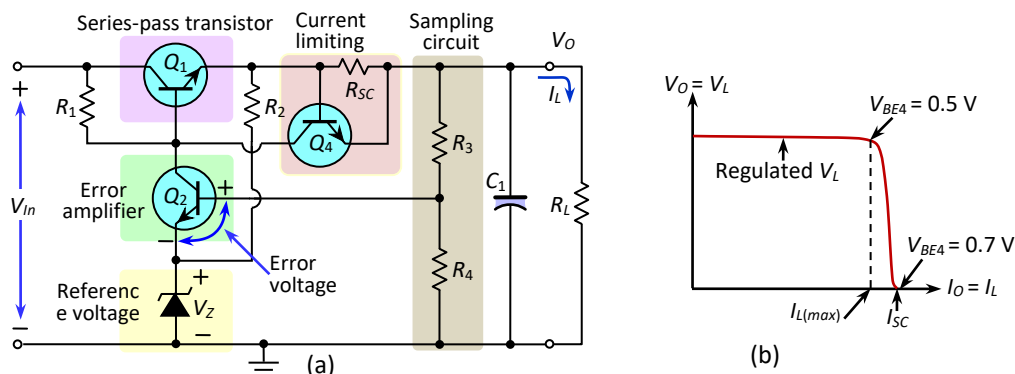


Fig.18-21: Series type voltage regulator with constant current limiting facility
(a) Schematic diagram, and (b) output characteristics

To make the concept clearer, let us consider $R_{SC} = 1 \Omega$ that will set $I_{L(max)} = 500 \text{ mA}$. Since the load current has to pass through R_{SC} , the current sensing resistor produces the base-emitter voltage for Q_1 . When the load current is less than or equal to 500 mA [$I_L \leq I_{L(max)}$], the voltage across R_{SC} is less than or equal to 0.5 V . In this case, Q_4 is cut-off and the regulator works as previously described. When the load current is greater than 500 mA (say $I_L = 600 \text{ mA}$) the voltage across R_{SC} will be 0.6 V . This voltage turns Q_4 in conduction. The collector current of Q_4 flows through R_1 . This increases the voltage drop across R_1 and decreases the base voltage to Q_1 , which reduces the load voltage and the load current.

When the output terminals are shorted, Q_4 conducts heavily and brings the base voltage of Q_1 down to approximately 1.4 V ($V_{B1} = V_{BE1} + V_{BE4}$). The current through the pass transistor is typically a little greater than $I_{L(max)}$.

Fig.18-18 summarizes the concept of current limiting. As an approximation, the graph shows 0.5 V as the voltage at which current limiting begins and 0.7 V as the voltage under short-circuit conditions. When the load current is small, the output voltage is regulated and has a value of $V_O = V_L$. When I_L increases, the load voltage remains constant up to a V_{BE} of, approximately, 0.5 V. Beyond this point, Q_4 turns ON and current limiting mechanism sets in. Further increases in I_L decrease the load voltage, and regulation does not work. When the load is shorted, the load current is limited to a value of I_{SC} , the load current with shorted-load terminals. When the load terminals are shorted in Fig. 22-18, the maximum load current I_{SL} that can be drawn from the circuit is the current necessary to create $V_{BE4} \approx 0.7$ V across R_{SC} . Mathematically,

$$I_{SC} = \frac{V_{BE4}}{R_{SC}} \quad (18-13)$$

where V_{BE4} can be approximated as 0.7 V. For heavy load currents, the V_{BE4} of the current-sensing transistor may be somewhat higher. We used an R_{SC} of 1 Ω in our discussion. By changing the value of R_{SC} , we can get current limiting to begin at any level. For instance, if $R_{SC} = 10 \Omega$, current limiting would start at approximately 60 mA with a shorted-load current of approximately 70 mA.

The overload and short-circuit protection circuit can also be developed using two diodes as shown in Fig.18-22. Here, the diodes will be forward biased by $V_{BE1} + V_{R_{SC}}$. Though $V_{BE1} = 0.7$ V, the voltage drop across R_{SC} will be less than 0.7 V for the rated maximum load current. Hence the diodes will not be forward biased. However, if the the load current exceeds the rated current or the output is shorted, $V_{R_{SC}}$ will be greater than 0.7 V. Therefore, $V_{BE1} + V_{R_{SC}}$ will forward bias the diodes and the current will flow through the short-circuit path as shown in Fig.18-22. Rest of the operation process is same as before.

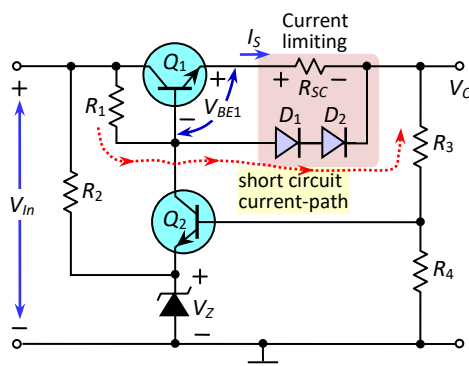


Fig.18-22: Voltage regulator with current limiting circuit made of diodes

18.8.4 Voltage Regulator with Foldback Current Limiting Circuit

Already we have discussed that the foldback current limiting circuit overcomes the problem of constant current limiting circuit. In constant current limiting circuits, if the output current exceeds the rated maximum current, the output voltage decreases with the increase in output current. So, most of the input voltage is dropped across the series-pass transistor and a large amount of current passes through it in overload or short-circuit condition. For this reason huge power is dissipated in this series-pass transistor. On the other hand, in case of foldback current limiting circuit, if the output current crosses the rated maximum current limit, both the output voltage and current decreases. Therefore, though the voltage drop across the series-pass transistor is high, the current through this transistor decreases (foldback), which reduces the power dissipation.

A voltage regulator circuit with the foldback current limiting mechanism is shown in Fig.18-23. Note that this circuit is almost similar to the constant current limiting circuit of Fig.18-22. The only

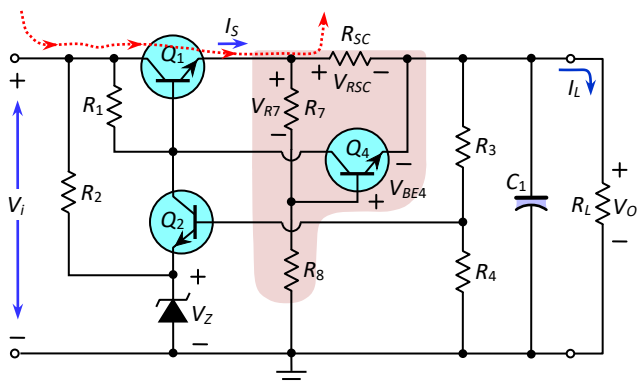


Fig.18-23: Block diagram of a series type voltage regulator with foldback current limiting mechanism

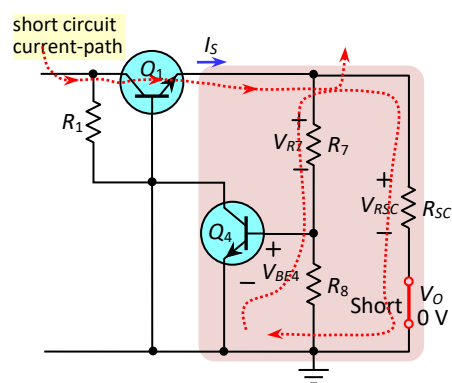


Fig.18-24: Current limiting circuit redrawn using short circuit

difference is a voltage divider network ($R_7 - R_8$) is used to bias the protection transistor Q_4 . To understand this circuit, let us assume that the output terminals are shorted. The short circuit condition is shown in Fig.18-24. For this circuit the voltage drop across the base-emitter junction of Q_4 can be determined using KVL around the loop shown in Fig.18- 24.

$$V_{BE4} + V_{R7} - V_{RSC} = 0$$

$$\text{or,} \quad \boxed{V_{BE4} = V_{RSC} - V_{R7}} \quad (18-14)$$

From Fig.18-23, it is clear that $V_{RSC} + V_o$ will be applied across the voltage divider network $R_7 - R_8$. So the voltage across R_7 can be calculated using voltage divider rule,

$$V_{R7} = \left(\frac{R_7}{R_7 + R_8} \right) (V_{RSC} + V_o) \quad (18-15)$$

For normal operation, V_{RSC} will be very small because the value of R_{SC} is very small. Therefore,

$$\boxed{V_{R7} \approx \left(\frac{R_7}{R_7 + R_8} \right) V_o = K V_o} \quad (18-16)$$

where,

$$K = \frac{R_7}{R_7 + R_8} \quad (18-17)$$

According to the Equ.(18-16), V_{R7} will increase or decrease if the output voltage (load voltage) increases or decreases but, the value of V_{RSC} will increase or decrease if the output current (load current) increases or decreases. When the load current increases to its maximum permissible value, V_{RSC} becomes large enough to make $V_{BE4} \approx 0.7$ V, that is, V_{RSC} becomes larger than V_{R7} by about 0.7 V. At this point, transistor Q_4 is forward biased and current limiting mechanism occurs, just as it does in the constant current limiting circuit of Fig.18-21. If the load resistance is now made smaller, the load voltage will drop, as we have seen previously. But according to Equ.(18-16), when the load voltage drops, V_{R7} also drops. Consequently, by Equ.(18-14), a smaller value of V_{RSC} is required to maintain $V_{BE4} = 0.7$ V. Since V_{BE4} remains constant at 0.7 V, a smaller load current

must flow to produce the smaller drop across R_{SC} . Further decrease in load resistance produces further drops in load voltage and a further reduction in load current, so foldback limiting occurs. If the load resistance is restored to its normal operating value, the circuit resumes normal regulator action.

For short-circuit condition ($V_o = 0$) the circuit can be redrawn as shown in Fig.18-24. From Equ.(18-15) we get,

$$V_{R7} = \left(\frac{R_7}{R_7 + R_8} \right) V_{RSC}$$

Therefore,
$$V_{BE4} = V_{R8} = \left(\frac{R_8}{R_7 + R_8} \right) V_{RSC} = (1 - K)V_{RSC}$$

But, Q_4 will turn ON for only $V_{BE4} = 0.7$ V. Therefore, the value of foldback current (or short-circuit current) at short-circuit condition will be,

$$0.7 \text{ V} = (1 - K)V_{RSC} = (1 - K)I_{SC}R_{SC}$$

$$\therefore \boxed{I_{SC} = \frac{0.7 \text{ V}}{(1 - K)R_{SC}}} \quad (18-18)$$

The permissible maximum load current can be calculated from Equ.(18-17) and Equ.(18-18).

Calculation of Foldback Factor:

For normal operation we can use Equ.(18-15),

$$V_{BE4} = V_{RSC} - V_{R7}$$

Using Equ.(18-15),
$$V_{BE4} = V_{RSC} - \left(\frac{R_7}{R_7 + R_8} \right) (V_{RSC} + V_o)$$

or,
$$V_{BE4} = I_L R_{SC} - \left(\frac{R_7}{R_7 + R_8} \right) (I_L R_{SC} + V_o)$$

or,
$$V_{BE4} = \frac{I_L R_{SC} R_7 + I_L R_{SC} R_8 - I_L R_{SC} R_7 - V_o R_7}{R_7 + R_8}$$

or,
$$V_{BE4}(R_7 + R_8) = I_L R_{SC} R_8 - V_o R_7$$

or,
$$I_L = \frac{V_{BE4}(R_7 + R_8)}{R_{SC} R_8} + \frac{V_o R_7}{R_{SC} R_8} \quad (18-19)$$

As $K = R_7/(R_7 + R_8)$, $\therefore (R_7 + R_8)/R_8 = 1/(1 - K)$, and $R_7/R_8 = K/(1 - K)$. Therefore, Equ.(18-19) can be written as,

$$I_L = \frac{V_{BE4}}{(1 - K)R_{SC}} + \frac{KV_o}{(1 - K)R_{SC}}$$

At the maximum load current $I_{L(max)}$, the value of V_{BE4} will be almost 0.7 V. Therefore,

$$I_{L(max)} = \frac{0.7 \text{ V}}{(1 - K)R_{SC}} + \frac{KV_o}{(1 - K)R_{SC}}$$

or,
$$I_{L(max)} = \frac{0.7 \text{ V} + KV_o}{(1 - K)R_{SC}} \quad (18-20)$$

Using Equ.(18-18), Equ.(18-20) can also be written as,

$$I_{L(max)} = I_{SC} + \frac{KV_o}{(1 - K)R_{SC}} \quad (18-21)$$

The performance of this foldback current limiting circuit is represented by a factor called **foldback factor** (or **foldback ration**) which is defined as the ration of maximum current to the short circuit current. Mathmatically,

$$\text{Foldback ratio} = \frac{I_{L(max)}}{I_{SC}} \quad (18-22)$$

Using Equ.(18-18) and Equ.(18-20) into Equ.(18-22),

$$\begin{aligned} \text{Foldback ratio} &= \frac{0.7 \text{ V} + KV_o}{(1 - K)R_{SC}} / \frac{0.7 \text{ V}}{(1 - K)R_{SC}} \\ \text{or,} \quad \text{Foldback ratio} &= \frac{0.7 \text{ V} + KV_o}{(1 - K)R_{SC}} \times \frac{(1 - K)R_{SC}}{0.7 \text{ V}} \end{aligned}$$

or,
$$\text{Foldback ratio} = 1 + \frac{KV_o}{0.7 \text{ V}} \quad (18-23)$$

Example 18-7

Suppose a foldback current limiting network is constructed using $R_{SC} = 2.7 \Omega$, $R_7 = 510 \Omega$, $R_8 = 7.5 \text{ k}$, $R_{SC} = 2.7 \Omega$. If the output voltage is $V_o = 12 \text{ V}$, calculate the rated maximum load current, the short circuit current and the foldback factor. Draw the response curve of this current limiting circuit.

Solution:

Using Equ.(18-17),
$$K = \frac{R_7}{R_7 + R_8} = \frac{510 \Omega}{510 \Omega + 7500 \Omega} \approx 0.0637$$

Using Equ.(18-20),
$$I_{L(max)} = \frac{0.7 \text{ V} + KV_o}{(1 - K)R_{SC}} = \frac{0.7 \text{ V} + 0.0637 \times 12 \text{ V}}{(1 - 0.0637) \times 2.7 \Omega}$$

or,
$$I_{L(max)} \approx 579 \text{ mA}$$

Using Equ.(18-18),
$$I_{SC} = \frac{0.7 \text{ V}}{(1 - K)R_{SC}} = \frac{0.7 \text{ V}}{(1 - 0.0637)2.7 \Omega}$$

or,
$$I_{SC} \approx 277 \text{ mA}$$

Using Equ.(18-23),

$$\text{Foldback ratio} = 1 + \frac{KV_o}{0.7 \text{ V}}$$

$$= 1 + \frac{0.0637 \times 12 \text{ V}}{0.7 \text{ V}} \approx 2.1$$

The characteristic curve is drawn in Fig.18-25.

Comments: By increasing the value of K , we can increase the foldback factor.

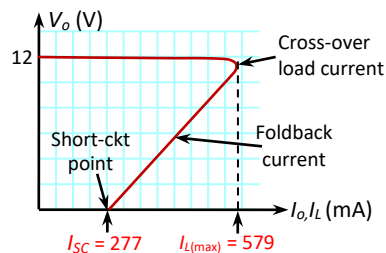


Fig.18-25: Output characteristics of the Example 18-7

18.9 Series Regulator using Op-Amp

Instead of using a BJT as the error amplifier, operational amplifier can also be used as an error amplifier. The high voltage gain and high input impedance of op-amp provide better performance of the regulator circuit. A simple series voltage regulator is shown in Fig.18-26. Compare this circuit with the series regulator consisting of BJTs. This circuit is similar to that of Fig.18-9. The only difference is, here the error amplifier is comprised of an operational amplifier (A_1). The constant reference voltage (V_Z) is applied to the noninverting input and the sampled voltage (V_{R3}) is applied to the inverting input terminal. The output sample circuit (R_2 and R_3) always senses the output voltage to detect any change. The error amplifier compares the sampled voltage with the reference voltage and drives the control transistor to compensate the change.

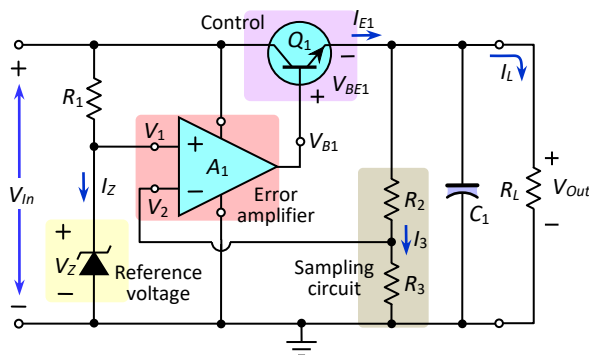


Fig.18-26: Series type voltage regulator using op-amp

The operation process of this regulator is same as described before. The voltage divider network formed by R_2 and R_3 senses any change in the output voltage. When the output tries to decrease, because of a decrease in input voltage (V_{in}) or because of an increase in load current caused by a decrease in R_L , the voltage at the op-amp's inverting terminal (V_2) decreases proportionately. Since the reference voltage at the op-amp's nonverting terminal ($V_1 = V_Z$) remains nearly constant, a small difference voltage (error voltage) is developed across the op-amp's inputs ($V_{id} = V_1 - V_2$). This difference voltage is amplified, and the op-amp's output voltage, V_{B1} , increases. This increase is applied by the op-amp [$V_{B1} = A_1(V_1 - V_2)$] to the base of Q_1 , causing the output voltage ($V_o = V_{B1} - V_{BE1}$) to increase until the voltage to the inverting input (V_2) again approximately equals the reference voltage (V_Z). This action compensates the attempted decrease in output voltage and thus keeps it nearly constant. Here also, the series-pass transistor, Q_1 , is usually placed on a heat sink because all of I_L must flow through it.

Exactly the opposite action occurs when the output voltage tries to increase due to the increase in V_{in} , or decrease in I_L . The op-amp in the series regulator is actually connected as a noninverting

amplifier where the reference voltage V_{RF} is the input at the noninverting terminal, and the voltage divider (R_2 and R_3) forms the negative feedback circuit (neglecting V_{BE1}). Thus the closed-loop voltage gain of the error amplifier will be,

$$A_{CL} = \left(1 + \frac{R_2}{R_3}\right) \quad (18-24)$$

Therefore, the regulated output voltage of the series regulator is,

$$V_o = \left(1 + \frac{R_2}{R_3}\right) V_{RF} - V_{BE1} = \left(1 + \frac{R_2}{R_3}\right) V_Z - V_{BE1} \quad (18-25)$$

Neglecting the base-emitter voltage of Q_1 ,

$$V_o = \left(1 + \frac{R_2}{R_3}\right) V_{RF} = \left(1 + \frac{R_2}{R_3}\right) V_Z \quad (18-26)$$

From this equation, we find that the output voltage is determined by the reference voltage (or Zener voltage) and the resistors (R_2 and R_3). It is relatively independent of the input voltage, and therefore, regulation is achieved (as long as the input voltage and load current are within specified limits). The load regulation and the source regulation of this circuit will be better because of very high open-loop gain of the op-amp [refer to Example 18-3].

Example 18-8

The series voltage regulator of Fig.18-27 has been designed for a nominal input voltage of 25 V. Calculate the output voltage. Calculate the output voltage for $\pm 20\%$ change in input voltage.

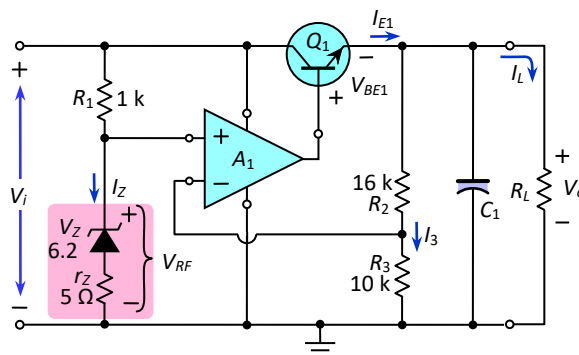


Fig.18-27: Series type voltage regulator using op-amp

Solution:

Neglecting the input current of the op-amp,

$$I_Z = \frac{V_i - V_Z}{R_1 + r_z} = \frac{25 \text{ V} - 6.2 \text{ V}}{1 \text{ k} + 5 \Omega} = 18.71 \text{ mA}$$

$$V_{RF} = V_Z + r_z I_Z = 6.2 \text{ V} + 18.71 \text{ mA} \times 5 \Omega = 6.294 \text{ V}$$

Using Equ.(18-25),
$$V_o = \left(1 + \frac{R_2}{R_3}\right) V_{RF} - V_{BE1} = \left(1 + \frac{16 \text{ k}}{10 \text{ k}}\right) \times 6.294 \text{ V} - 0.7 \text{ V} \approx 15.66 \text{ V}$$

For 20% increase in input voltage, $V_i = 25 \text{ V} + 25 \text{ V} \times 20/100 = 30 \text{ V}$.

$$\therefore I_Z = \frac{V_i - V_Z}{R_1 + r_z} = \frac{30 \text{ V} - 6.2 \text{ V}}{1 \text{ k} + 5 \Omega} = 23.68 \text{ mA}$$

$$\therefore V_{RF} = V_Z + r_z I_Z = 6.2 \text{ V} + 23.68 \text{ mA} \times 5 \Omega = 6.318 \text{ V}$$

$$\text{Using Equ.(18-25), } V_0 = \left(1 + \frac{16 \text{ k}}{10 \text{ k}}\right) \times 6.318 \text{ V} - 0.7 \text{ V} \approx 15.73 \text{ V}$$

Considering 20% decrease in input voltage, $V_i = 25 \text{ V} - 25 \text{ V} \times 20/100 = 20 \text{ V}$.

$$\therefore I_Z = \frac{V_i - V_Z}{R_1 + r_z} = \frac{20 \text{ V} - 6.2 \text{ V}}{1 \text{ k} + 5 \Omega} = 13.73 \text{ mA}$$

$$\therefore V_{RF} = V_Z + r_z I_Z = 6.2 \text{ V} + 13.73 \text{ mA} \times 5 \Omega = 6.269 \text{ V}$$

$$\text{Using Equ.(18-25), } V_0 = \left(1 + \frac{16 \text{ k}}{10 \text{ k}}\right) \times 6.269 \text{ V} - 0.7 \text{ V} \approx 15.6 \text{ V}$$

Results: The circuit has been simulated in LTspice and the results are shown in Fig.18-. The simulated results are little bit higher than calculated due to higher Zener voltage (6.4 V).

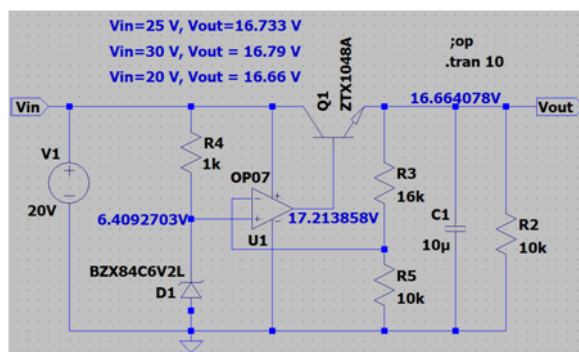


Fig.18-28: The circuit of Example 18-8 simulated in LTspice

Comments: For a change of $\pm 20\%$ in the input voltage the regulated output voltage has been changed by $(15.73 \text{ V} - 15.6 \text{ V})/15.6 \text{ V} \approx 0.83\%$.

18.9.1 High Currnt op-amp Volatge Regulator

As discussed before (Fig.18-17), we can use Darlington pair to increase the output current capacity of a voltage regulator circuit. The output current capacity of Fig.18-26 can also be improved by using Darlington pair. The circuit arrangement is shown in Fig.18-29. The operation process of this regulator will be same as before.

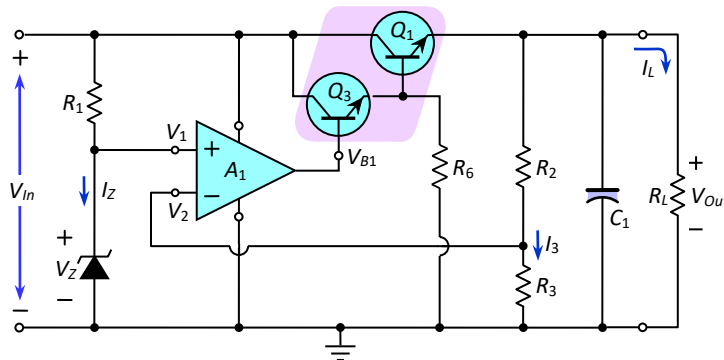


Fig.18-29: Series type voltage regulator using op-amp and Darlington pair

18.9.2 Op-amp Voltage Regulator with Variable Output

Like the BJT voltage regulator op-amp voltage regulator can be designed to produce variable output voltage. The circuit arrangement is shown in Fig.18-30. The operation process of this regulator will be same as that of Fig.18-12.

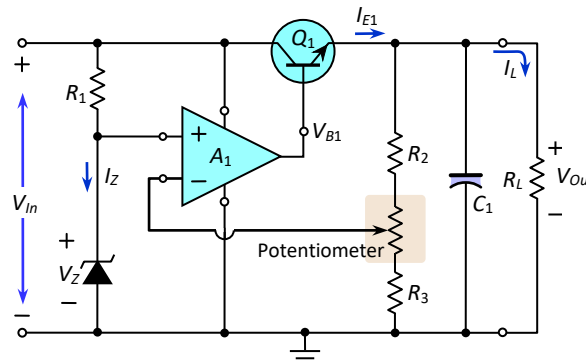


Fig.18-30: Op-amp voltage regulator with variable output

18.10 Current Liming Circuit with Op-amp Voltage Regulator

We have already discussed that the voltage regulators with high current capacity must include an overload (or short-circuit) protection mechanism. In section, two types of protection circuits have been discussed. These protection circuits may also be used in op-amp voltage regulators. Fig.18- shows an op-amp voltage regulator with a constant current limiting circuit. Here Q_1 is Darlington power transistor that provides high current capacity of this circuit. If we compare this circuit with that of Fig.18-, we find an additional resistor R_4 has been used inbetween the op-amp's output and the common point of base of series-pass transistor and the collector of current limiting transistor (Q_2). The working process of this current limiting circuit is same as described in Section . We will understand the necessity of R_4 . Due to overload or short-circuit the output voltage decreases. So the voltage at the inverting terminal (V_{R3}) will decrease and hence the output of the op-amp will increase. For short circuit condition, the output will be almost zero. So V_{R3} will also be zero. Therefore, the op-amp's output will be very high almost equal to the input voltage V_i . For

short circuit condition transistor Q_2 will be ON and work as short circuit. As the internal output resistance of the op-amp is very small, a very large current will flow if the resistance R_4 were not used. For this reason R_4 is used to reduce the current flowing through the op-amp and the transistor Q_2 . In fact R_4 will provide the gain of the CE amplifier consisting of Q_2 . The value of R_4 should be large enough to provide sufficient gain but it should not be so large that the op-amp cannot drive the series-pass transistor (Q_1) in normal operation. Typically the value of R_4 resistor can be few ohms to few hundred ohm.

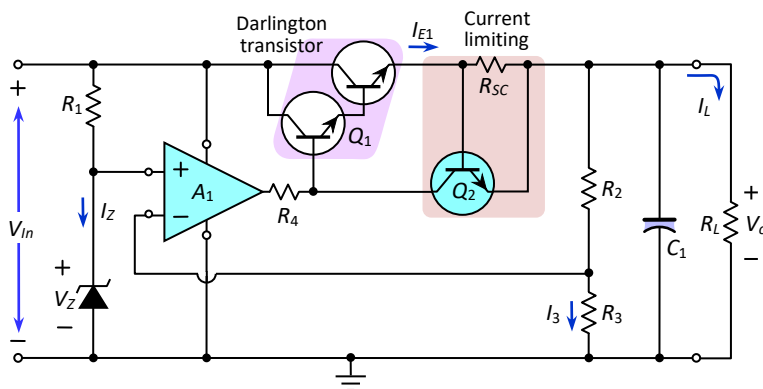


Fig.18-31: Series type voltage regulator using op-amp

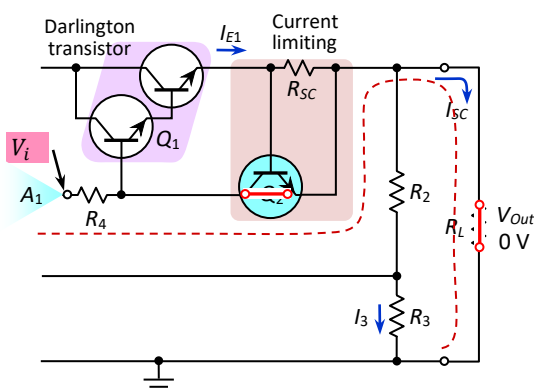


Fig.18-32: Series type voltage regulator using op-amp

For normal operation, that is, for $I_L \leq I_{L(max)}$, the maximum base current of Q_1 will be $I_{L(max)}/h_{FE1}$. $R_4 = V_i/I_{C2}$.

Example 18-9

Design a voltage regulator circuit like Fig.18-31 with the nominal output voltage of $12\text{ V} \pm 20\%$ and the maximum output current $I_{L(max)} = 1.2\text{ A}$. The supply or input voltage is $V_i = 20\text{ V}$. The Darlington transistor has $h_{FE1} = 450$, and $V_{BE1} = 1.4\text{ V}$ and that of the current limiting transistor is $h_{FE2} = 150$. Assume the bias current of the op-amp is $I_B = 100\text{ nA}$ and the internal output resistance is $r_o = 75\ \Omega$.

Solution:

Nominal output voltage $V_o = 12 \text{ V}$

Minimum output voltage $V_{o(min)} = 12 \text{ V} - 12 \text{ V} \times 0.2 = 9.4 \text{ V}$

Maximum output voltage $V_{o(max)} = 12 \text{ V} + 12 \text{ V} \times 0.2 = 14.4 \text{ V}$

Let $V_Z = 0.5 \times V_o = 0.5 \times 12 \text{ V} = 6 \text{ V}$. So we can use 1N4735A that has $V_Z = 6.2 \text{ V}$, knee Zener current $I_{ZK} = 1 \text{ mA}$ and the maximum Zener current $I_{ZM} = 146 \text{ mA}$.

The design Zener current (I_Z) should be such a value that $I_Z \gg I_B$ (op-amp's biasing current) and $I_{ZK} < I_Z < I_{ZM}$. To satisfy these conditions let us select $I_Z = 10 \text{ mA}$.

The value of R_1 can be calculated as,

$$R_1 = \frac{V_i - V_Z}{I_Z} = \frac{20 \text{ V} - 6.2 \text{ V}}{10 \text{ mA}} \approx 1.4 \text{ k} \text{ [use standard } 1.5 \text{ k]}$$

The current in the sampling network should be much greater than I_B . Therefore, let $I_3 = 1 \text{ mA}$. As the value of I_B is very small, we can neglect this current. So, $I_2 = I_3 = I_5$.

To adjust the output voltage the potentiometer R_3 has been added. When the wiper of this potentiometer is placed at the topmost position, the output of the regulator circuit will be the minimum. The required minimum input is, $V_{o(min)} = 12 \text{ V} - 12 \text{ V} \times 0.2 = 9.4 \text{ V}$.

As the open-loop gain of the op-amp is very high, it is reasonable to assume $v_1 = v_2 = 6.2 \text{ V}$. Therefore,

$$V_{R2} = V_o - v_2 = 9.4 \text{ V} - 6.2 = 3.2 \text{ V}$$

$$\text{or,} \quad R_2 = \frac{3.2 \text{ V}}{I_2} = \frac{3.2 \text{ V}}{1 \text{ mA}} = 3.2 \text{ k} \text{ [use standard } 3.3 \text{ k]}$$

$$\text{and,} \quad V_{R3} + V_{R4} = v_2 = 6.2 \text{ V}$$

$$\therefore \quad R_3 + R_4 = \frac{6.2 \text{ V}}{1 \text{ mA}} = 6.2 \text{ k}$$

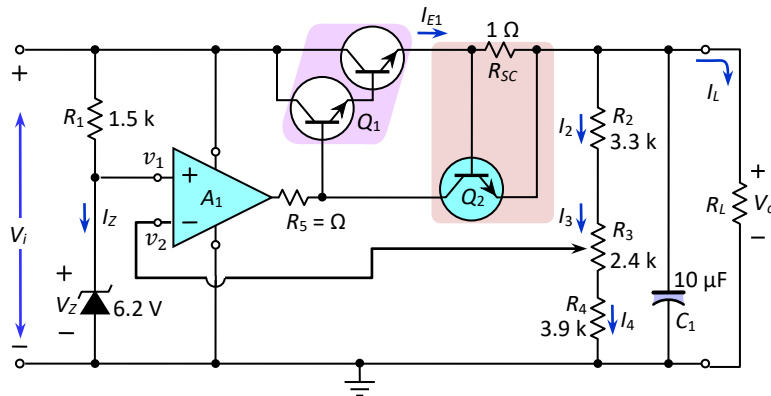


Fig.18-33: Op-amp voltage regulator of Example 18-9

Similarly, when the wiper is placed at the bottom-most position, the output voltage will be the maximum, $V_{o(max)} = 14.4 \text{ V}$. Therefore, the current through this branch will be,

$$I_2 = \frac{V_{o(max)}}{R_2 + R_3 + R_4} = \frac{14.4 \text{ V}}{3.3 \text{ k} + 6.2 \text{ k}} \approx 1.52 \text{ mA} = I_4$$

But, $V_{R4} = v_2 = V_Z$

$$\text{or, } R_4 = \frac{V_Z}{I_4} = \frac{6.2 \text{ V}}{1.52 \text{ mA}} \approx 4.08 \text{ k} [\text{use standard } 3.9 \text{ k}]$$

Now, $R_3 + R_4 = 6.2 \text{ k}$

or, $R_3 = 6.2 \text{ k} - R_4$

or, $R_3 = 6.2 \text{ k} - 3.9 \text{ k} = 2.3 \text{ k} [\text{use standard } 2.4 \text{ k pot.}]$

Neglecting the current in the sampling circuit we get $I_{sc} = I_{L(max)} = 1.2 \text{ A}$.

$$\text{Using Equ.(18-13), } R_{SC} = \frac{V_{BE4}}{I_{SC}} = \frac{0.7 \text{ V}}{1.2 \text{ A}} \approx 1 \Omega$$

The maximum base current of Q_1 will be,

$$I_{B1(max)} = \frac{I_{L(max)}}{h_{FE1(min)}} = \frac{1.2 \text{ A}}{450} \approx 2.67 \text{ mA}$$

Let $I_{C2(max)} = I_{B1(max)} = 2.67 \text{ mA}$

$$\therefore R_5 = \frac{V_o + V_{BE1}}{I_{C2(max)} + I_{B1(max)}} = \frac{12 \text{ V} + 1.4 \text{ V}}{2.67 \text{ mA} + 2.67 \text{ mA}} \approx$$

18.11 Shunt Voltage Regulator

In a shunt voltage regulator, the controlling element (BJT or other device) is connected in parallel with the load that shunts the current to ground. The shunt regulator maintains a constant voltage across its terminals by adjusting the shunt current flowing through it. Shunt regulators are inefficient because maximum current is drawn from the source regardless of the load current, i.e. even when there is no load current (output open).

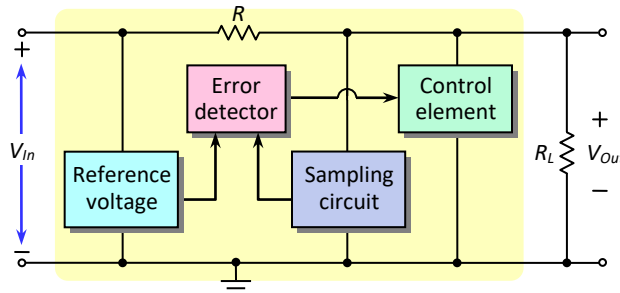


Fig.18-34: Block diagram of a shunt type voltage regulator

The basic functional components of a shunt voltage regulator are shown in the block diagram of Fig.18-34. The operation of each block is same as that of series regulator. In series regulator the output voltage is maintained constant by adjusting the voltage across the controlling element. Whereas, in the shunt regulator, the voltage drop across R_1 is adjusted to maintain a constant voltage across the control element. The Zener diode voltage regulator, described in Chapter 2, is the example of basic shunt voltage regulator. The circuit is redrawn in Fig.18-35.

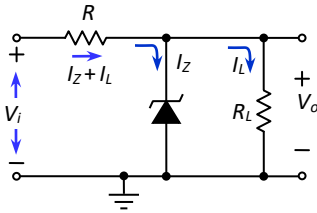


Fig.18-35: Shunt type voltage regulator using Zener diode

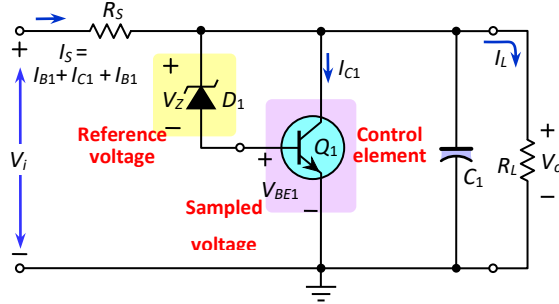


Fig.18-36: Shunt type voltage regulator using BJT

18.11.1 Shunt Thpe Voltage Regulator Using BJT

A shunt voltage regulator using discrete components is shown in Fig.18-36. Here the transistor Q_1 connected in parallel (in shunt) with the load is the controlling element and the Zener diode provide the reference voltage. The Zener diode will also work as the sampling circuit and the error voltage, $V_{BE1} = V_o - V_Z$ will drive the control element (Q_1). The above figure shows the circuit diagram of transistor shunt regulator where the transistor is connected in shunt (parallel) with the load hence the circuit is called transistor shunt regulator.

When the output voltage (V_o) tries to decrease due to a derease in input voltage (V_{In}) or due to increase in load current (caused by a decrease in load resistance), the attempted decrease decrease V_{BE1} , and hence the collector of Q_1 decrease that reduces the voltage drop across R_{SE} . Therefore, the output voltage increases to its original value.

Similarly, when V_o tries to increase due to an increase in V_{In} or due to decrease in load current, the attempted decrease will also decrease V_{BE1} . So the collector current I_{C1} will decrease and hence the current through the resistor R_{SE} will decrease and the output voltage will increase to its previous value.

In this way any attempt to change the output voltage will also change V_{BE1} in oppsit direction and the this change is amplified by Q_1 and the voltage drop across R_{SE} is adjusted to keep the output voltage almost constnt.

It may be noted from the circuit that the output voltage V_o is equal to the sum of zener voltage and the base to emitter voltage V_{BE1} of transistor Q_1 ,

$$V_o = V_Z + V_{BE1} \quad (18-27)$$

18.11.2 Shunt Voltage Regulator Using Op-Amp

The schematic diagram of a shunt voltage regulator using op-amp is shown in Fig.18-37. Here, the reference voltage (V_{RF}) is made by the Zener diode (D_1), the error amplifier is made by the op-amp

(A_1) and the sampling network is made of the voltage divider (R_2 and R_3). The controlling element is the shunt transistor Q_1 .

The operation of this circuit is similar to that of the series regulator discussed so far, the only difference is that the regulation is achieved by changing the voltage drop across R_1 by controlling the current through the parallel transistor Q_1 .

When the output voltage (V_o) tries to decrease due to a decrease in input voltage (V_i) or due to increase in load current (caused by a decrease in load resistance), the attempted decrease is sensed by R_3 and R_4 and applied to the op-amp's noninverting input. Thus the voltage in the noninverting input of the operational amplifier will try to decrease.

As the voltage at the inverting terminal of the op-amp remains constant (V_{RF}), the output voltage of the op-amp decreases. As a result the shunt current (I_{C1}) through transistor Q_1 decreases. Therefore, the voltage drop across the resistor R_1 decreases and output voltage increases ($V_o = V_i - V_{R1}$) to its original value.

Similarly, when V_o tries to increase due to an increase in V_{in} or due to decrease in load current, the attempted increase is sensed by R_3 and R_4 and applied to the op-amp's noninverting input. So the output voltage of the op-amp increases. As a result I_{C1} of Q_1 increases. Therefore, the voltage drop across the resistor R_1 increases and output voltage decreases ($V_o = V_i - V_{R1}$) to keep it nearly constant.

The total current flowing through the series resistor (R_S) will be $I_S = I_{C1} + I_2 + I_L \approx I_{C1} + I_L$. With I_L and V_o constant, a change in the input voltage (ΔV_i) produces a change in this shunt current (I_S) as follows,

$$\Delta I_S = \frac{\Delta V_i}{R_S} \quad (18-28)$$

With a constant V_i and V_o , a change in load current (ΔI_L) causes an opposite change in shunt current by the same amount. If I_L increases, I_S decreases, and vice versa.

$$\Delta I_S = -\Delta I_L \quad (18-29)$$

The main disadvantage of the shunt regulator is its inefficiency but the main advantage is that it has inherent short circuit protection. If the output is shorted ($V_o = 0$ V), the maximum short-circuit current will be limited by the series resistor R_S . The value of this current will be,

$$I_{S(max)} = \frac{V_{i(max)}}{R_S} \quad (18-30)$$

The power rating of this series resistor should be,

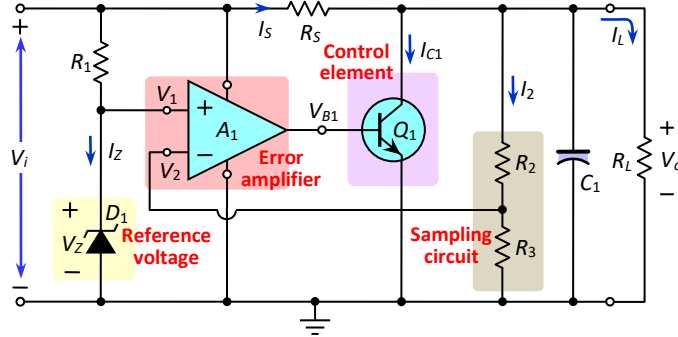


Fig.18-37: Shunt type voltage regulator using op-amp

$$P_{R_S} = \frac{V_{In(max)}^2}{R_S} = I_{S(max)}^2 R_S \quad (18-31)$$

As $I_{S(max)}$ is significantly large, the required power capacity of R_S should be large. Practically a resistor with power capacity greater than that found by Equ.(18-31) has to be used.

Example 18-10

Design a voltage regulator circuit like Fig.18-37 with the nominal output voltage of 12 V and the maximum output current $I_{L(max)} = 200$ mA. The range of supply voltage is $V_{In} = 16$ V to 20 V. Assume the bias current of the op-amp is $I_B = 100$ nA and the transistor has $h_{FE1} = 200$.

Solution:

Nominal output voltage $V_o = 12$ V

Let $V_Z = 0.5 \times V_o = 0.5 \times 12$ V = 6 V and use 1N4735A that has $V_Z = 6.2$ V. As described in Example 18-, let us select $I_Z = 5$ mA for $V_i = 16$ V.

The value of R_1 can be calculated as,

$$R_1 = \frac{V_{In} - V_Z}{I_Z} = \frac{16 \text{ V} - 6.2 \text{ V}}{5 \text{ mA}} \approx 2 \text{ k}$$

The current in the sampling network should be much greater than the biasing current of the op-amp I_B . Therefore, let $I_2 = 1$ mA. As the value of I_B is very small, we can neglect this current. So, $I_2 = I_3$.

$$v_1 = v_2 = V_Z = V_{R3} = I_3 R_3$$

$$R_3 = \frac{V_Z}{I_3} = \frac{6.2 \text{ V}}{1 \text{ mA}} = 6.2 \text{ k}$$

and,

$$V_{R2} = I_2 R_2 = V_o - V_{R3}$$

and,

$$R_2 = \frac{V_o - V_{R3}}{I_2} = \frac{12 \text{ V} - 6.2 \text{ V}}{1 \text{ mA}} = 5.8 \text{ k} \text{ [use standard } 5.6 \text{ k]}$$

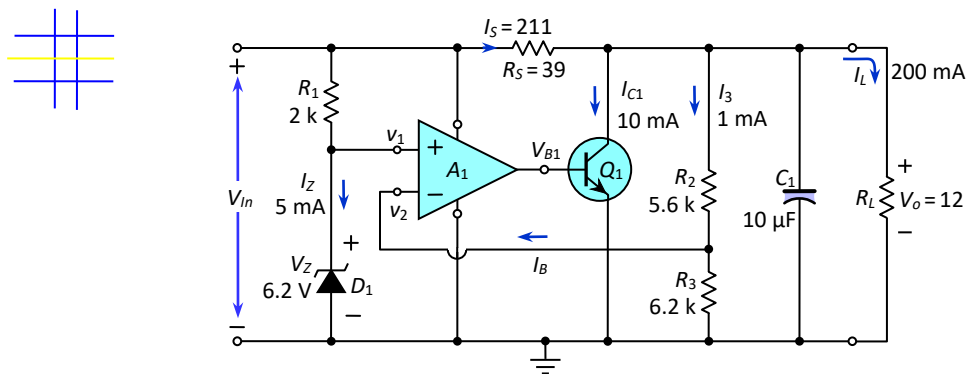


Fig.18-38: Series type voltage regulator using op-amp

The value of R_S has to be calculated using the $I_{L(max)}$ and $V_{In(max)}$. Considering $I_{C1} = 10$ mA,

$$R_S = \frac{V_{In} - V_o}{I_{C1} + I_3 + I_L}$$

$$R_S = \frac{20\text{ V} - 12\text{ V}}{10\text{ mA} + 1\text{ mA} + 200\text{ mA}} \approx 38\ \Omega \text{ [use standard } 39\ \Omega]$$

Power rating of this transistor should be,

$$P_{Rs} = I_{S(max)}^2 R_S = (210\text{ mA})^2 \times 39\ \Omega \approx 2\text{ W}$$

Results: The designed circuit is shown in Fig.18-38.

Comments: At open circuit condition $I_{C1} = 210$ mA, and $V_{CE1} = 12$ V. Therefore, the transistor specification must satisfy these requirements.

18.12 Three-Terminal Series Voltage Regulator ICs

Different voltage regulator ICs are available in the market. For example, the 78xx is a family of self-contained fixed linear voltage regulator integrated circuits (ICs). The 78xx family has been commonly used in electronic circuits for many years because of their ease-of-use, low cost, and excellent performance. Not only the 78xx series that give positive output voltage, but there are also the complementary 79xx series that give negative voltage output. For both the series, xx is a two digit number that represents the value of output regulated voltage. For example, L7805 (or LM7805, MC7805 etc.) is linear voltage regulator IC that produce +5 V, and L7905 (or LM7905, MC7905 etc.) is the complementary ICs that gives -5 V output. Some other common IC numbers are shown in Table 18-1 and 18-2.

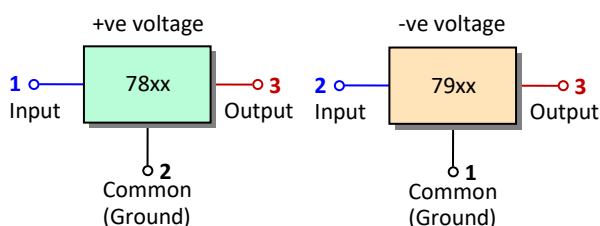


Fig.18-39: 78xx and 79xx ICs pin-outs

The main package type of the 78xx and 79xx series regulators is the TO220. However, they are also available in some other packages as shown in Fig.18-. All of these voltage regulator ICs have only three terminals (pin outs). For this reason they are called **three-terminal voltage regulators**. The pin-out is very simple: (1) input, (2) common (ground), and (3) output. Pin-diagrams of different packages are shown in Fig.18-. The metal on the packages (TO-220, TO-220FP, and TO-3) is connected to common (ground) terminal, so this is ideal for mounting onto heat sinks which are normally mechanically and electrically connected to the ground of the system. The current capacity of these packages is 1 A. But the current capacity of smaller package (TO-92, SOT-89) is only 200 mA.

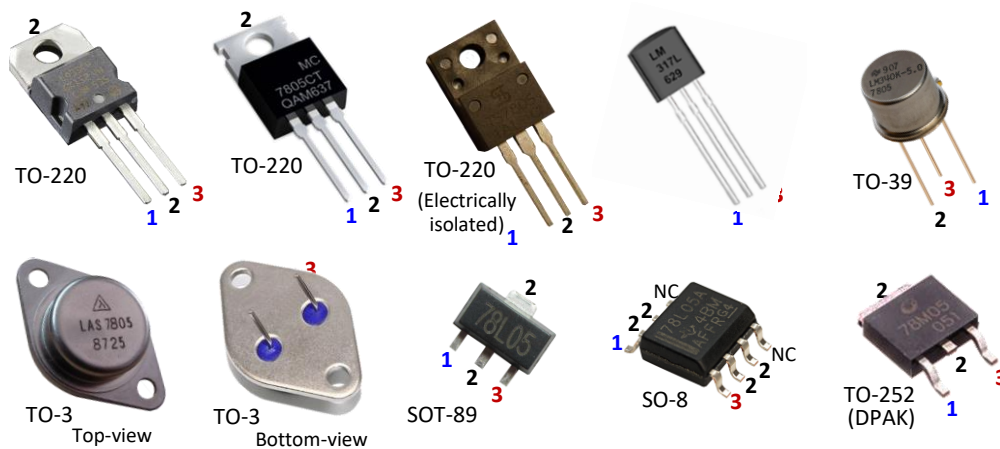


Fig.18-40: Different packages of 78xx and 79xx ICs

18.12.1 Block Diagram of 78xx ICs

The internal block diagram of 78xx ICs is illustrated in Fig.18-41. The **starting circuit** provides initial current to the reference circuit, so it doesn't get stuck in an OFF state. The **reference voltage** unit is a band-gap type reference voltage generating circuit. It is a temperature independent voltage reference circuit (produced by using the band-gap voltage of B-E junction of a transistor) widely used in integrated circuits. It produces a fixed (constant) voltage regardless of power supply variations, temperature changes, or circuit over loading. For proper functioning of the reference voltage unit, the **current generator** unit produces a constant current. The voltage divider, consisting of two resistors, scales down the output voltage and applies to the **error amplifier** unit. The error amplifier compares this scaled-down voltage with the reference voltage and amplifies the error voltage to control the **series-pass element**. The series-pass element may be a Darlington pair transistor (or some other power elements) that controls the output voltage and current.

For example, in a 7805 IC the output voltage will be 5 V. For this 5 V output, the voltage divider provides 3.75 V and supplies this voltage into the error amplifier. For any change in output voltage, the voltage divider will produce more or less than 3.75 V. The error amplifier amplifies the difference and control the series-pass element, so that the output voltage comes back to 5 V again.

The **thermal protection** unit provides protection against overheating of the series-pass element due to excessive input voltage, and excessive output current. If internal power dissipation becomes too high for the heat sinking provided, the thermal protection circuit shuts down the IC, and hence, protect from overheating and self- destruction.

For any power semiconductor device (like BJT, MOSFET, IGBT etc.), the **safe operating area** (SOA) is defined as the voltage and current conditions over which the device can be expected to operate without self-damage. The purpose of the **SOA protection** unit is to control the current through and or voltage across the series-pass element to save it from self- destruction when the output is short.

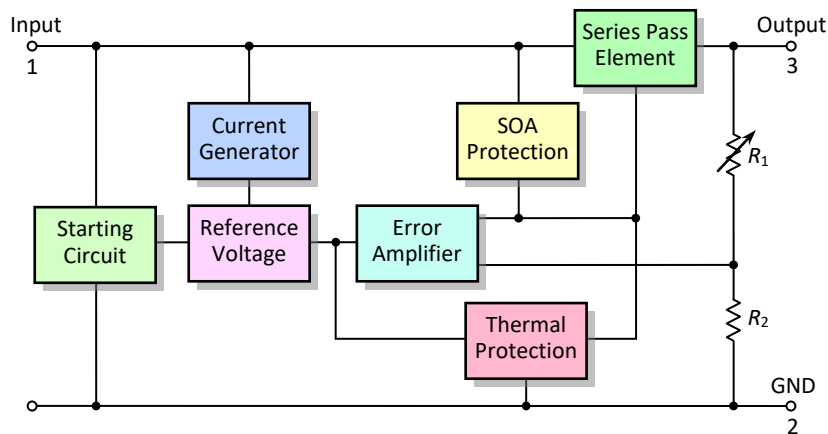


Fig.18-41: Internal block diagram of 78xx ICs

18.12.2 78xx Datasheet and Specifications

Different semiconductor manufacturing companies produce 78xx voltage regulator ICs. Some of the renowned manufacturing companies are - National Semiconductor, Fairchild Semiconductor, Texas Semiconductor etc. A portion of the datasheet of 78xx ICs of Fairchild Semiconductor Corporation is given below. The 78xx series contains ICs of different voltages: 5, 6, 8, 9, 10, 12, 15, 18, 24 V. The specifications only for the 5 V IC (LM7805) are given here.

The typical load regulation is 9 mV for a load current between 5 mA and 1.5 A. The typical line regulation is 4 mV for an input voltage of 7 to 25 V. It also has a ripple rejection of 73 dB, which means that it will reduce the input ripple by a factor of 4,467. With an output resistance of approximately 15 m Ω , the LM7805 is a very stiff voltage source to all loads within its current rating.

The input voltage range and the output voltage of some other ICs are given in Table 18-1, and Table 18-2.

Table 18-1: Some positive voltage regulator ICs

IC Number	Nominal Output Voltage (V)	Input Voltage Range (V)	Maximum Output Current (mA)
7805	5.0 V	7 V to 20 V	1000 mA
7806	6.0 V	8 V to 21 V	1000 mA
7809	9.0 V	11.5 V to 24 V	1000 mA
7812	12.0 V	14.5 V to 27 V	1000 mA
7815	15.0 V	17.5 V to 30 V	1000 mA
7818	18.0 V	21 V to 33 V	1000 mA
7824	24.0 V	27 V to 38 V	1000 mA



LM78XX / LM78XXA

3-Terminal 1 A Positive Voltage Regulator

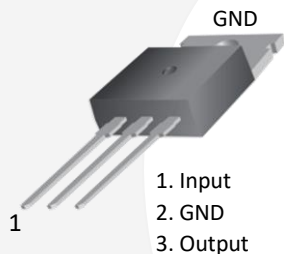
Features

- Output Current up to 1 A
- Output Voltages: 5, 6, 8, 9, 10, 12, 15, 18, 24 V
- Thermal Overload Protection
- Short-Circuit Protection
- Output Transistor Safe Operating Area Protection

Description

The LM78XX series of three-terminal positive regulators is available in the TO-220 package and with several fixed output voltages, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shut-down, and safe operating area protection. If adequate heat sinking is provided, they can deliver over 1 A output current. Although designed primarily as fixed-voltage regulators, these devices can be used with external components for adjustable voltages and currents.

TO-220 (Single Gauge)



Block Diagram

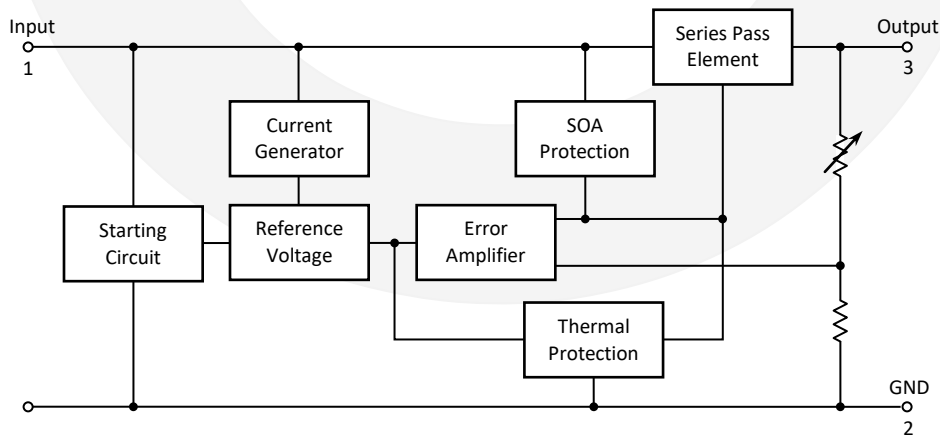


Figure 1. Block Diagram

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at $T_A = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value	Unit
V_I	Input Voltage	$V_O = 5\text{ V to }18\text{ V}$	V
		$V_O = 24\text{ V}$	
$R_{\theta JC}$	Thermal Resistance, Junction-Case (TO220)	5	$^\circ\text{C/W}$
$R_{\theta JA}$	Thermal Resistance, Junction-Air (TO220)	65	$^\circ\text{C/W}$
T_{OPR}	Operating Temperature Range	LM78xx	$^\circ\text{C}$
		LM78xxA	
T_{STG}	Storage Temperature Range	-65 to +150	$^\circ\text{C}$

Electrical Characteristics (LM7805)

(Refer to the test circuit, $-40^\circ\text{C} < T_J < 120^\circ\text{C}$, $I_O = 500\text{ mA}$, $V_I = 10\text{ V}$, $C_1 = 0.1\text{ }\mu\text{F}$, unless otherwise specified)

Symbol	Parameter	Min.	Typ.	Max.	Unit
V_O	Output Voltage	$T_J = +25^\circ\text{C}$	4.80	5.00	5.20
		$I_O = 5\text{ mA to }1\text{ A}$, $P_O \leq 15\text{ W}$, $V_I = 7\text{ V to }20\text{ V}$	4.75	5.00	5.25
Regline	Line Regulation ⁽²⁾	$T_J = +25^\circ\text{C}$	$V_I = 7\text{ V to }25\text{ V}$	4.0	100.0
			$V_I = 8\text{ V to }12\text{ V}$	1.6	50.0
Regload	Load Regulation ⁽²⁾	$T_J = +25^\circ\text{C}$	$I_O = 5\text{ mA to }1.5\text{ A}$	9.0	100.0
			$I_O = 250\text{ mA to }750\text{ mA}$	4.0	50.0
I_Q	Quiescent Current	$T_J = +25^\circ\text{C}$	5	8	mA
ΔI_Q	Quiescent Current Change	$I_O = 5\text{ mA to }1\text{ A}$	0.03	0.50	mA
		$V_I = 7\text{ V to }25\text{ V}$	0.30	1.30	
$\Delta V_O / \Delta T$	Output voltage Drift ⁽³⁾	$I_O = 5\text{ mA}$	-0.8		mV/ $^\circ\text{C}$
V_N	Output Noise Voltage	$f = 10\text{ Hz to }100\text{ kHz}$, $T_A = +25^\circ\text{C}$	42		μV
RR	Ripple Rejection ⁽³⁾	$f = 120\text{ Hz}$, $V_I = 8\text{ V to }18\text{ V}$	62	73	dB
V_{DROP}	Dropout Voltage	$T_J = +25^\circ\text{C}$, $I_O = 1\text{ A}$	2		V
R_O	Output Resistance ⁽³⁾	$f = 1\text{ kHz}$	15		m Ω
I_{SC}	Short-Circuit Current	$T_J = +25^\circ\text{C}$, $V_I = 35\text{ V}$	230		mA
I_{PK}	Peak Current ⁽³⁾	$T_J = +25^\circ\text{C}$	2.2		A

Notes:

- Load and line regulation are specified at constant junction temperature. Changes in V_O due to heating effect must be taken into account separately. Pulse testing with low duty is used.
- These parameters, although guaranteed, are not 100% tested in production.

Table 18-2: Some negative voltage regulator ICs

IC Number	Nominal Output Voltage (V)	Input Voltage Range (V)	Maximum Output Current (mA)
7905	-5.0 V	-7 V to -20 V	1000 mA
7906	-6.0 V	-9 V to -21 V	1000 mA
7909	-9.0 V	-11.5 V to -23 V	1000 mA
7912	-12.0 V	15.5 V to -27 V	1000 mA
7915	-15.0 V	17.5 V to 30 V	1000 mA
7918	-18.0 V	-18 V to -30 V	1000 mA
7924	-24.0 V	22.5 V to 33 V	1000 mA

18.12.3 Application of 78xx

78xx three-terminal linear voltage regulator can be used to produce constant voltage of different output voltage with 1 A current capacity. If adequate heat sinking is provided, they can deliver over 1.5 A output current. Though they are manufactured to provide constant output voltage, they can also be used to produce variable output voltage to some extent. Some important and common applications are discussed in the following headings.

Fixed Output Voltage Regulators

LM78xx and LM79xx series ICs can be very easily used to produce a constant output voltage. As the ICs have only three terminals, the circuit configuration is very simple as shown in Fig.18-42. Here, the 'xx' represents the value of output voltage. Thus if we want to produce regulated 5 V, then we have to use a LM7805 IC. Similarly if want to produce regulated -5 V, then we have to use a LM7905 IC. Or, to produce regulated 12 V (or, -12 V) output, we have to use LM7812 (or, LM7812) has to be used. Remember that the input voltage must be typically 2.0 V higher than the output voltage and the maximum output voltage is given in Table 1 and 2.

When an IC is more than 6 inches from the filter capacitor of the unregulated power supply, the inductance of the connecting wire may produce oscillations inside the IC. This is why manufacturers recommend using a bypass capacitor C_i on pin 1 (Fig. 22-18b). On the other hand, the output capacitor, C_o , is used to improve stability and transient response. Note that C_i and C_o are much smaller than required in discrete component power supply. A large reservoir capacitor is not necessary, as the regulating action of the IC will reduce the amplitude of any AC ripple (within its maximum input voltage range) to just a few millivolts at the output. Typical values for either bypass capacitor are from 0.1 to 1 F. The data sheet of the 78xx series suggests using a 0.33 μ F or larger tantalum, mylar, or other capacitor with low internal impedance at high frequencies for the input capacitor and 0.1 F for the output capacitor.

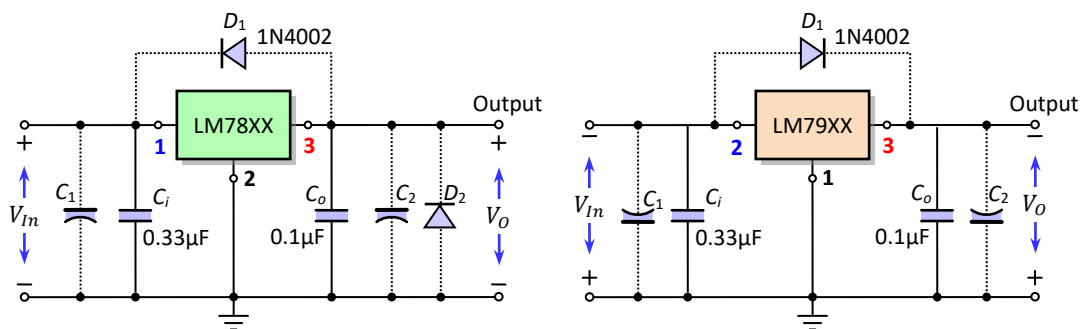


Fig.18-42: Internal block diagram of 78xx ICs

In some circuits electrolytic capacitors may be used for C_i and C_o as an alternative to using tantalum or polyester capacitors, but in this case the capacitance should be considerably large, 25 μF or more. In some circuits a 0.1 μF low impedance capacitor is used to remove transients or spikes and a 100 μF electrolyte capacitor is used for decoupling purpose in both input and output sides. In circuits where C_o is 100 μF or more, there is a possibility that if the input is shorted to ground, either temporarily (or permanently because of a fault) that the charge on C_o will cause a large current to flow back into the IC output terminal that may damage the IC. To prevent this, a diode such as a 1N4002 can be used across the IC as shown in Fig.18-42 so that if, at any time the input terminal is at a lower potential than the output, the diode will conduct any charge at the output terminal to ground, rather than allowing the current to flow through the IC. D_1 also provides reverse bias protection in the input side. Similarly, D_2 is used to protect the ICs from output polarity reversal.

Effect of Open GND Connection on a 78xx IC

If, for any reason, the ground connection to the IC is disconnected momentarily as indicated in Fig.18-43, the output voltage may rise to the unregulated input voltage, which could cause damage to components being supplied by the regulator. This may happen if a 78xx IC is plugged into an IC socket with supply voltage ON.

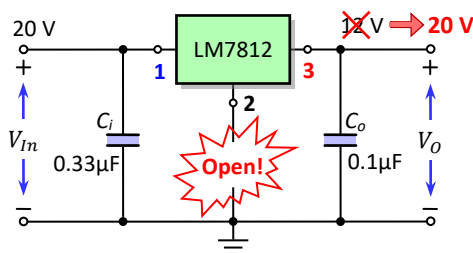


Fig.18-43: LM7812 IC with GND open

18.12.4 Voltage Regulator IC as Constant Current Source

Voltage regulator ICs can be used as a constant current source as shown in Fig.18-44. 78XX ICs produce constant output voltage so long as the input voltage and the output current are in their specified tolerance range. Therefore, if a resistor (R_1) is connected at the output terminals of the voltage regulator circuit, a constant current $I_1 = XX \text{ V}/R_1$ will flow through this resistor. So, the total current flowing through the load resistor will be,

$$I_o = I_Q + I_1 = I_Q + \frac{XX V}{R_1} \quad (18-32)$$

Here, I_Q is the quiescent (biasing) current of the 78XX IC and its value is given in the datasheets and XX is the regulated output voltage.

For example, if we use a LM7805 IC with $R_1 = 5 \Omega$, the circuit will work as a 1.0 A current source (neglecting I_Q). The power dissipation in the resistor will be $P_{R1} = 5 \text{ W}$.

18.12.5 Voltage Regulator IC with High Current Capacity and Current Limiting

The maximum output current capacity of 78xx voltage regulator is 1 A. In some application this current capacity may not be sufficient. But voltage regulator with higher current capacity can be implemented using extra transistor with 78xx voltage regulator. A circuit arrangement is shown in Fig.18-. Here, transistor Q_1 is high current capacity power transistor and Q_2 is a normal transistor used for short-circuit protection purpose. The value of resistor R will be so selected that when the current through the 78xx IC exceeds 1 A transistor Q_1 will be forward biased and the additional current will flow through this current. Similarly, the value of R_{SC} can be calculated as $R_{SC} = V_{BE2}/I_{SC}$.

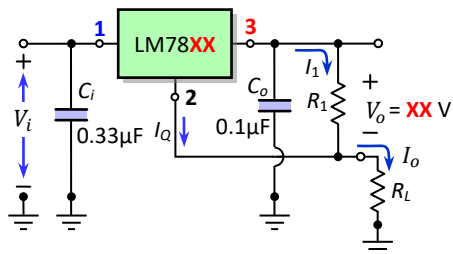


Fig.18-44: LM78XX IC as constant current source

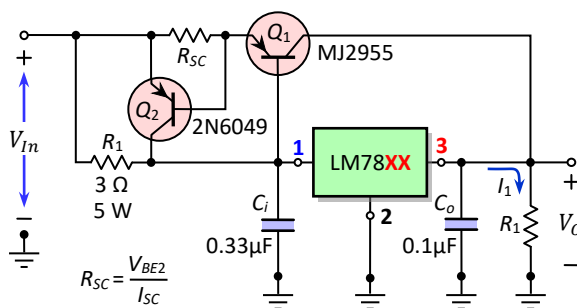


Fig.18-45: High output current with short-circuit protection

18.13 LM317 and LM337 Variable Voltage Regulators IC

The LM117 (positive) and LM337 (negative) series of adjustable 3-pin voltage regulators are capable of supplying 1.5 A with an output voltage range of 1.25 V to 37 V and a wide temperature range. They are very easy to use and require only two external resistors to set the output voltage. Further, both line and load regulation are better than standard fixed regulators. Like LM78xx, and LM79xx, these ICs are available in different packages. Fig. shows the pin-outs of only the TO220 package.

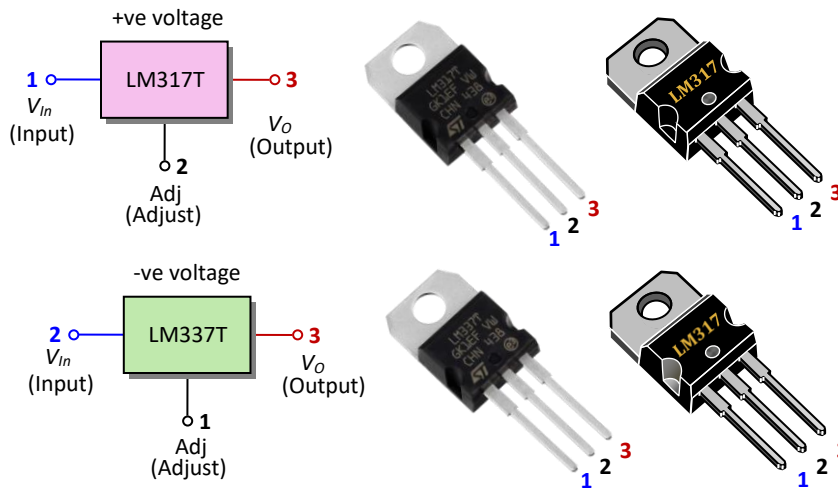


Fig.18-46: LM317T and LM337T pin-outs

The LM117 and LM337 offer full protection such as current limit, thermal overload protection and safe area protection. All overload protection circuitry remains fully functional even if the adjustment terminal is disconnected.

Typically, no capacitors are needed unless the device is situated more than 6 inches from the input filter capacitors, in which case an input bypass is needed. An optional output capacitor can be added to improve transient response. The adjustment terminal can be bypassed to achieve very high ripple rejection ratios that are difficult to achieve with standard 3-terminal regulators.

Because the regulator is floating and detects only the input-to-output differential voltage, supplies of several hundred volts can be regulated as long as the maximum input-to-output differential is not exceeded. That is, avoid short-circuiting the output.

By connecting a fixed resistor between the adjustment pin and output, the LM117 and LM337 can be also used as a precision current regulator. Supplies with electronic shutdown can be achieved by clamping the adjustment terminal to ground, which programs the output to 1.25 V where most loads draw little current.

For applications requiring greater output current, see the LM150 series (3 A) and LM138 series (5 A) data sheets. For the negative complement, see the LM137 series data sheet.

Some specifications of this ICs are given below,

- 1.5-A Output Current
- Output Adjustable between 1.2 V and 37 V
- Typical Line Regulation 0.01%/V
- Typical Load Regulation 0.3%
- Ripple Rejection 77 dB
- Thermal Overload Protection
- Internal Short-Circuit Current Limiting and Protections

18.13.1 Internal Block Diagram of LM317

Like other voltage regulator ICs, the schematic diagram of LM317 is very large and complicated. But to understand the operation mechanism, the functional block diagram is shown in Fig.18-47. As shown in this figure, the constant current source and the Zener diode develop a constant voltage of 1.25 V which is applied to the non-inverting terminal of the error amplifier. The error amplifier is consisted of an operation amplifier. The input voltage to the non-inverting input of this error amplifier can be adjusted using an external voltage divider network. The output voltage is directly connected to the inverting terminal of the error amplifier. The error amplifier controls a Darlington connected series-pass transistors. Like other voltage regulator ICs, the over temperature and over current protection mechanism is implemented internally.

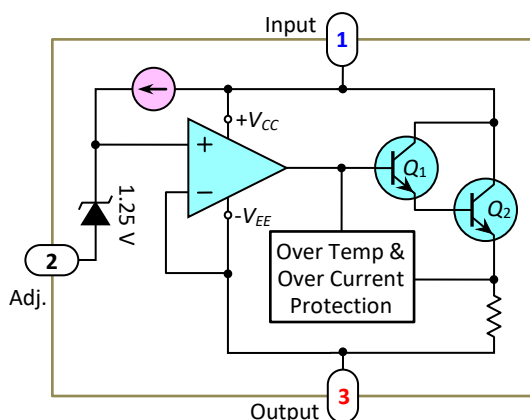


Fig.18-47: Internal block diagram of LM317

18.13.2 Adjustable Voltage Regulator using LM317 and LM337

The LM317 and LM337 are versatile, high performance, linear regulators with high accuracy and a wide temperature range. Its functionality can be utilized in many different applications that require high performance regulation, such as battery chargers, constant current regulators, and microprocessor supplies.

Fig.18-48 shows how LM317 can be used as a variable voltage regulator. The LM317 is a 3-terminal floating regulator. The LM317 develops and maintains a nominal 1.25 V reference between its output (pin-3) and adjustment terminal (pin-2). The input side of the circuit of Fig.18-48, has been redrawn in Fig.18-49 showing the internal components. The differential input voltage

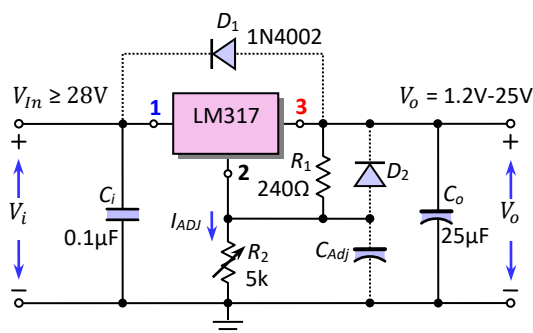
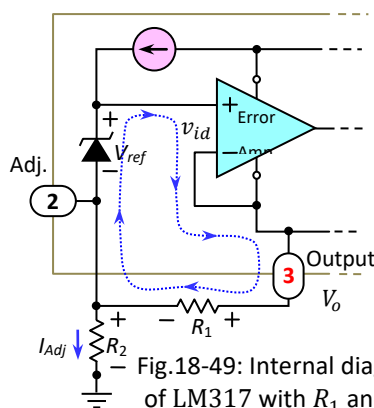


Fig.18-48: Variable positive voltage regulator circuit

Fig.18-49: Internal diagram of LM317 with R_1 and R_2

at the error amplifier (v_{id}) will be zero as the open-loop gain of the op-amp is very large. As the output voltage is divided by R_1 and R_2 , The output across R_2 is given by,

$$V_{R1} = V_o \left(\frac{R_1}{R_1 + R_2} \right) \quad (18-33)$$

Applying KVL around the loop shown in Fig.18 we get,

$$-V_{R1} + V_{Ref} - v_{id} = 0 \quad (18-34)$$

But $v_{id} \approx 0$, therefore, putting the value of V_{R1} into the above equation,

$$V_o \left(\frac{R_1}{R_1 + R_2} \right) = V_{Ref}$$

or,

$$V_o = V_{Ref} \left(1 + \frac{R_2}{R_1} \right)$$

But as shown in Fig.18-49, due to the current from the adjustment terminal (I_{Adj}), a very small voltage will be added to the output. Therefore, the total output voltage will be,

$$V_o = V_{ref} \left(1 + \frac{R_2}{R_1} \right) + I_{Adj} R_2 \quad (18-35)$$

Since I_{Adj} represents an error term in the equation, the LM317 was designed to control I_{Adj} to less than 100 μA and keep it constant.

A 0.1 μF ceramic disc or 1.0 μF tantalum input bypass capacitor (C_i) is recommended to reduce the sensitivity to input line impedance. The adjustment terminal may be bypassed to ground to improve ripple rejection. This capacitor (C_{Adj}) prevents ripple from being amplified as the output voltage is increased. The value of C_{Adj} should be 10 μF or above. Although the LM317 is stable with no output capacitance, like any feedback circuit, certain values of external capacitance can cause excessive ringing. A 1.0 μF tantalum or 25 μF aluminum electrolytic capacitor (C_o) on the output swamps this effect and ensures stability.

When external capacitors are used with any IC regulator it is sometimes necessary to add protection diodes to prevent the capacitors from discharging through the regulator IC. Figure 18 shows the LM317 with the recommended protection diodes (D_1 and D_2) for output voltages in excess of 25 V or high capacitance values ($C_o > 25 \mu F$, $C_{Adj} > 10 \mu F$). Diode D_1 prevents C_o from discharging thru the IC during an input short circuit. Diode D_2 protects against capacitor C_{Adj} discharging through the IC during an output short circuit. The combination of diodes D_1 and D_2 prevents C_{Adj} from discharging through the IC during an input short circuit.

If we need negative variable regulated supply voltage, we have to use LM337 IC. The circuit diagram for negative voltage is shown in Fig.18-50. The operation of this circuit is exactly same as the positive voltage regulator LM317. Note that D_1 has been connected in opposite direction

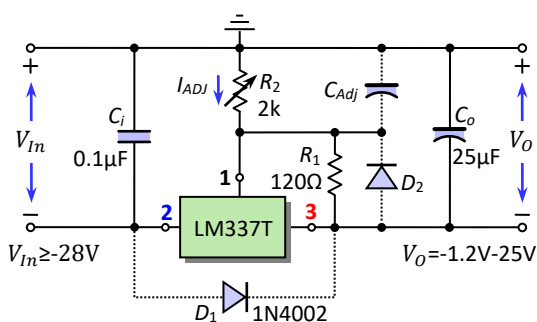


Fig.18-50: Variable negative voltage regulator circuit

compared to Fig.18-48 and the pin-1 and pin-2 of the IC are interchanged. All other connections remain exactly same. The output voltage of this circuit is also calculated by Equ. (18-35).

18.14 A Complete Variable Dual Power Supply Using LM317 & LM337

A complete variable dual power supply constructed using LM317 & LM337 is shown in Fig.18-51. The output voltage of this power supply is 0 to ± 30 V, and the maximum output current is 1.5 A. A 24 V centre-tape step down transformer is used to convert the AC 220 V to 24 V. The rectifier

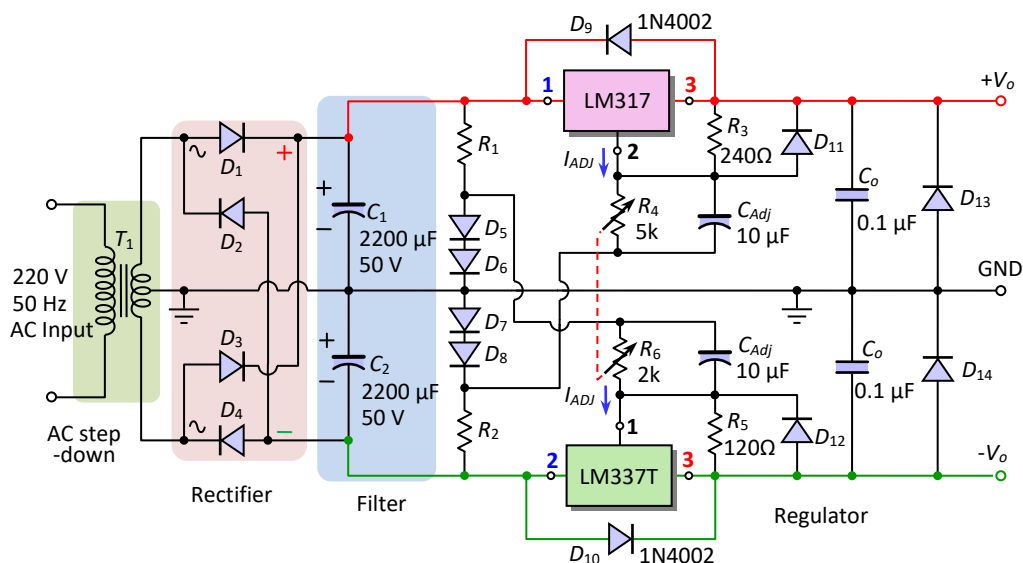


Fig.18-51: Circuit diagram of a complete variable dual power supply

consisting of diodes D_1 to D_4 converts this AC voltage into pulsating DC voltage. The centre-tape of the transformer will work as the ground (zero voltage) of this circuit. The positive output terminal of the bridge rectifier will work as the positive terminal (red line), and the negative output terminal of the bridge will work as the negative terminal (green line). To convert the pulsating DC into pure DC two capacitor filters have been used, one for positive side and the for negative side. After filtering the output voltage will be $\pm 22 \times \sqrt{2} \text{ V} = \pm 34 \text{ V}$ which is 4 V higher than the maximum output voltage. This unregulated positive and negative voltages have been regulated using LM317 and LM337 circuits of Fig.18-51.

In this circuit the variable resistors R_4 and R_6 are used to adjust the output voltage. These variable resistors should be mechanically coupled so that they can be changed by rotating a single knob (like a stereo type potentiometer). The only difference in the positive and negative voltage regulator circuits are the diodes D_5 to

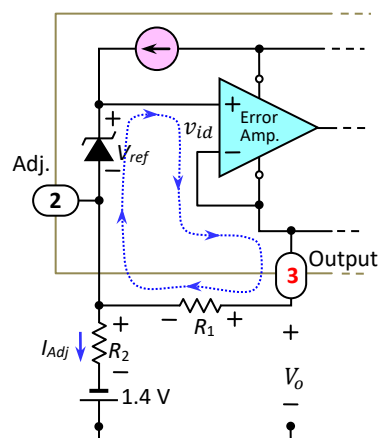


Fig.18-52: Internal diagram of LM317 with R_1 and R_2 and 1.4 V for diodes the diode voltage

D_8 . If we carefully observe the circuit, we find that these diodes will make a constant ± 1.4 V source. To have a better understanding, the input portion of Fig.18-52 is redrawn using a 1.4 V battery (generated by two diodes) considering Fig.18-49. From this circuit Equ.(18-33) can be written as,

$$V_{R1} = (+V_o + 1.4 \text{ V}) \left(\frac{R_1}{R_1 + R_2} \right) \quad (18-36)$$

As described before, we can write,

$$(+V_o + 1.4 \text{ V}) \left(\frac{R_1}{R_1 + R_2} \right) = V_{Ref}$$

or,

$$+V_o = V_{Ref} \left(1 + \frac{R_2}{R_1} \right) - 1.4 \text{ V}$$

Considering the current from the adjustment terminal (I_{Adj}),

$$+V_o = V_{ref} \left(1 + \frac{R_2}{R_1} \right) + I_{Adj} R_2 - 1.4 \text{ V} \quad (18-37)$$

Similarly, for the negative regulator,

$$-V_o = -V_{ref} \left(1 + \frac{R_2}{R_1} \right) - I_{Adj} R_2 + 1.4 \text{ V} \quad (18-38)$$

From these equations, we find that due to the diodes, the output voltage will be always 1.4 V less than the voltage generated by the ICs. Thus the output voltage range will be 0 to ± 30 V.

18.15 Voltage Regulator IC LM723

To design fixed regulated power supply we have discussed 78xx (positive voltage) and 79xx (negative voltage) ICs. On the other hand, we have discussed LM317 (positive voltage), and LM317 (negative voltage) voltage regulators that are used for variable power supply. Now, we will discuss Voltage Regulator IC LM723. This is one of the most popular IC used for the voltage regulation.



Fig.18-53: Pin configuration of LM723 IC

The LM723 is a monolithic integrated adjustable voltage regulator, assembled in a 14-pin dual-in-line plastic and SO-14 micro package, as shown in Fig.18-. The circuit provides internal current limiting. Provisions are made for adjustable current limiting and remote shut-down. It is also used

in various applications like shunt regulator, current regulator. The IC having low standby current drain, which allow us to use the IC as linear or fold-back current limiting, with an operating temperature ranges from -55°C to 150°C . Some specifications of this IC are given below,

- Series voltage regulator that provides positive and negative voltage
- Can be used as either a linear or a switching regulator.
- Provides current up to 150 mA, but with external transistor can provide up to 10 A
- Low standby current, very low-temperature drift, and high ripple rejection
- Input voltage range is 9 V to 40 V
- Output voltage range is 2 V to 37 V
- Load and line regulation is 0.01 %

Because of the small size and low cost, this IC is used in a wide variety of applications. It can be used as a series, shunt, linear, switching and floating regulators. Its wide voltage range, low noise, and extreme flexibility make it highly suitable for use in series voltage regulation applications.

18.15.1 Internal Block Diagram of LM723

The internal block diagram of LM723 is shown in Fig.18-54. The IC consists of mainly four functional units. The first unit is the **reference voltage generator** consisting of a temperature compensated Zener diode, a constant current source, and a reference voltage amplifier. The amplifier gives a very stable reference voltage of 7.15 V. The second unit is the **error amplifier** (or a comparator). As the reference voltage is not internally connected to the error amplifier, we have to connect this voltage to the noninverting input of the error amplifier externally by shorting pin-4 and pin-5. Similarly, a portion of the output voltage (divided by resistors) has to be connected to the inverting input of this error amplifier by connecting pin-1 and pin-2. With these external connections, the error amplifier compares the output voltage with the reference voltage and produces an error signal.

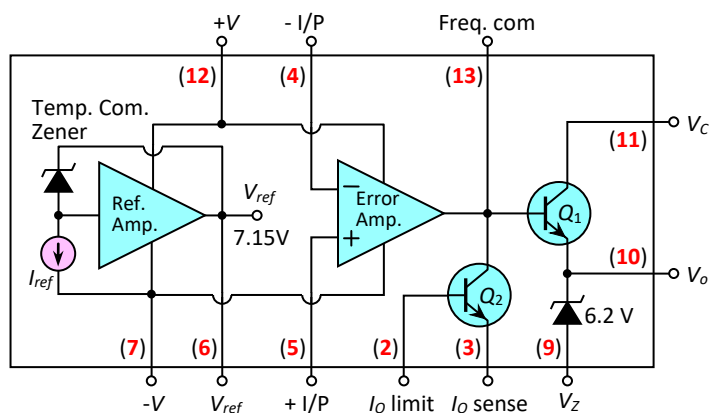


Fig.18-54: Block diagram of a regulated power supply

This error signal controls the third unit, the **series-pass transistor** Q_1 . The fourth unit of this regulator IC is the **current limiting transistor** (Q_2). To sense the output current an external resistor has to be attached between pin-2 and 3. Depending on the value of this resistor, the maximum output current of this IC will be set. If the output current is less than $I_{L(max)}$ transistor Q_2 is OFF, but if the output current exceeds the limit ($I_{L(max)}$), Q_2 turns ON and reduces the base current of the series-pass transistor (Q_1). Thus the output voltage decreases. In addition of these four units, the IC includes a Zener diode (D_2) for output voltage dropping in some applications.

18.15.2 Data Sheet of LM723



LM723, LM 723C

www.ti.com

SNVS765C –JUNE 1999–REVISED APRIL 2013



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

ABSOLUTE MAXIMUM RATINGS⁽¹⁾⁽²⁾ February 2014

Continuous Voltage from V^+ to V^-		40 V
Maximum Amplifier Input Voltage (Either Input)		8.5 V
Maximum Amplifier Input Voltage (Differential)		5 V
Current from V_Z		25 mA
Current from V_{REF}		15 mA
Internal Power Dissipation	PDIP	660 mW
	CDIP	900 mW
Operating Temperature Range	LM723	-55°C to +150°C
	LM723C	0°C to +150°C

Electrical Characteristics

Parameter	Conditions	LM723			LM723C			Units
		Min	Typ	Max	Min	Typ	Max	
Line Regulation	$V_{IN} = 12V$ to $V_{IN} = 15V$		0.01	0.1		0.01	0.1	% V_{OUT}
	$-55^\circ C \leq T_A \leq +125^\circ C$			0.3				
	$0^\circ C \leq T_A \leq +70^\circ C$						0.3	
	$V_{IN} = 12V$ to $V_{IN} = 40V$		0.02	0.2		0.1	0.5	
Load Regulation	$I_L = 1$ mA to $I_L = 50$ mA		0.03	0.15		0.03	0.2	% V_{OUT}
	$-55^\circ C \leq T_A \leq +125^\circ C$			0.6				
	$0^\circ C \leq T_A \leq +70^\circ C$						0.6	
Ripple Rejection	$f = 50$ Hz to 10 kHz, $C_{REF} = 0$		74			74		dB
	$f = 50$ Hz to 10 kHz, $C_{REF} = 5\mu F$		86			86		
Short Circuit Current Limit	$R_{SC} = 10\Omega$, $V_{OUT} = 0$		65			65		mA
Reference Voltage		6.95	7.15	7.35	6.80	7.15	7.5	V
Standby Current	$I_L = 0$, $V_{IN} = 30$ V		1.7	3.5		1.7	4.0	mA
Inout voltage range		9.5		40	9.5		40	V
Output oltage range		2.0		37	2.0		37	V
θ_{JA}	PDIP					105		°C/W
θ_{JA}	CDIP		150					°C/W
θ_{JC}			22			22		°C/W

18.15.3 Application of LM723 IC

LM723 is a versatile voltage regulator IC and can be used as fixed voltage and adjustable positive and negative voltage regulator. The current capacity of the IC is low (maximum 150 mA), but high output current (greater than 150 mA) can be produced by using external transistor. Similarly, the IC can be configured for either low voltage (2 V to 7 V) or 7 V to 37 V output voltage. Depending on this current capacity, and output voltage range, LM723 IC can be employed in four types of applications: (i) **Low voltage low current** (ii) **Low voltage high current** (iii) **High voltage low current**, and (iv) **High voltage high current**.

Application of LM723 IC as Fixed Voltage Regulator

The circuit diagram of a fixed positive voltage regulator is shown in Fig.18-55. This is a low voltage low current circuit. The unregulated input voltage is applied to pin-12 of LM723 to get a constant and regulated output voltage at pin-3. This voltage is then fed back through a resistor R_3 to the non-inverting pin of the IC (pin-4). The reference voltage at pin-6 (V_{ref}) is divided by a resistive network (R_1 and R_2) and applied to the non-inverting terminal (pin-5). The two voltages at the inverting and non-inverting input of the error amplifier are compared and an error signal is produced. This error signal controls the series pass transistor (Q_1) and keeps the output constant. The input voltage to the non-inverting terminal (pin-5) will be $V_{ref} \times R_2 / (R_1 + R_2)$. As the open-loop gain of the error amplifier is very large, it will try to make the voltage at the inverting terminal equal to that of noninverting terminal, that is, $V_{ref} \times R_2 / (R_1 + R_2)$. As input current of the error amplifier is very small, the voltage drop across R_3 will be very small and can be neglected. So output voltage (V_o) will be equal to the voltage at inverting terminal. Therefore, we can write the output voltage,

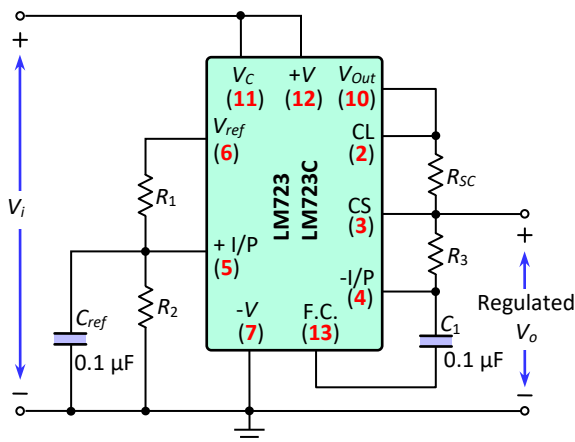


Fig.18-55: Regulated power supply using LM723

$$V_o = V_{ref} \times \frac{R_2}{R_1 + R_2} \quad (18-39)$$

As shown in Fig.18-54, value of V_{ref} is 7.15 V. Therefore, the maximum output voltage of this circuit will be approximately 7.5 V.

The function of the resistor R_3 is to minimize offset voltage by providing symmetrical biasing for both inverting and non-inverting terminal of the error amplifier. Therefore, the value of R_3 should be,

$$R_3 = \frac{R_1 R_2}{R_1 + R_2} \quad (18-40)$$

The function of the capacitor C_1 is to prevent the regulator circuit from oscillation or bypasses any high-frequency noise.

To adjust the output voltage, we can replace R_1 with a variable resistor (or potentiometer). By varying the resistance of that resistor, we can achieve the desired output voltage level. But the only one problem with this circuit is low output current, that is, $I_{o(max)} = 150$ mA.

Here R_{SC} is current sensing resistor that will limit the output current. As the output current of this circuit is 150 mA, the value of R_{SC} should be,

$$R_{SC} = \frac{V_{BE2}}{I_{o(max)}} = \frac{0.65 \text{ V}}{150 \text{ mA}} \approx 4.3 \Omega \quad (18-41)$$

If we need more than 7 V, we can arrange the circuit as shown in Fig.18- (High voltage low current). The only difference is the output voltage is divided by a resistive network (R_1 and R_2) and a fraction of output has been applied to the inverting input of error amplifier (pin-4). The voltage to the inverting input terminal will be,

$$V_o \times \frac{R_2}{R_1 + R_2}$$

As the open-loop gain of the error amplifier is very high, the voltages at the two input terminals should be same. Therefore,

$$V_o \times \frac{R_2}{R_1 + R_2} = V_{ref}$$

$$\text{or,} \quad V_o = V_{ref} \times \frac{R_1 + R_2}{R_2} \quad (18-42)$$

From this Equ.(18-), we find that the output voltage will be higher than $V_{ref} = 7.15$ V. Practically this circuit is able to produce 7 V to 37 V regulated output depending on the values of (R_1 and R_2).

As mentioned before, the function of R_3 is to provide symmetrical biasing condition in both input terminals of the error amplifier to minimize the temperature drift. The value of R_3 should be same as given by Equ.(18-40).

$$R_3 = \frac{R_1 R_2}{R_1 + R_2} \quad (18-43)$$

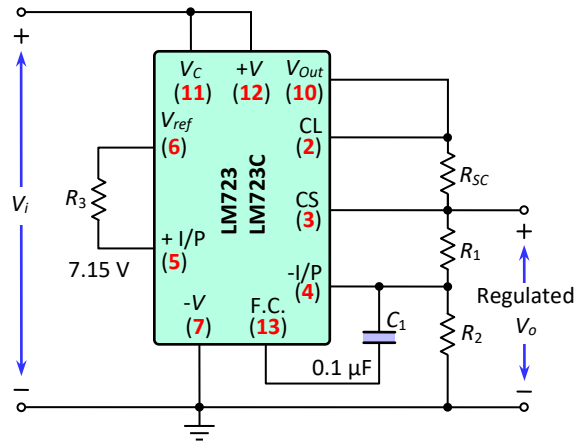


Fig.18-56: Regulated power supply using LM723 for $V_o > 7$ V

Example 18-11

Design a voltage regulator to produce $V_o = 5$ V, and $I_{o(max)} = 100$ mA using LM723 IC.

Solution:

The design requirements can be fulfilled using a low voltage low current configuration as shown in Fig.18-55 redrawn in Fig.18-57.

Using Equ.(18-39),
$$V_o = V_{ref} \times \frac{R_2}{R_1 + R_2} = 7.15 \text{ V} \times \frac{R_2}{R_1 + R_2}$$

The current through the voltage divider network (I_2) should be very large compared to the input bias current of the (internal) error amplifier which is typically 500 nA. Let us select $I_2 = 1 \text{ mA}$. To get $V_o = 5 \text{ V}$, the voltage at the inverting terminal (pin-5) should be 5 V. Therefore, from the Fig.18-57,

$$5 \text{ V} = V_{R2} = I_2 R_2 = 1 \text{ mA} \times R_2$$

$$\text{or, } R_2 = \frac{5 \text{ V}}{1 \text{ mA}} = 5 \text{ k} \text{ [use standard } 5.1 \text{ k]}$$

As the bias current is very small, we can assume $I_1 \approx I_2$. Therefore,

$$V_{R1} = V_{ref} - V_{R2} = 7.15 \text{ V} - 5 \text{ V} = 2.15 \text{ V}$$

$$\text{or, } I_1 R_1 = 7.15 \text{ V} - 5 \text{ V} = 2.15 \text{ V}$$

$$\therefore R_1 = \frac{2.15 \text{ V}}{1 \text{ mA}} = 2.15 \text{ k} \text{ [use standard } 2.2 \text{ k]}$$

Using Equ.(18-40),

$$R_3 = \frac{R_1 R_2}{R_1 + R_2} = \frac{2.2 \text{ k} \times 5.1 \text{ k}}{2.2 \text{ k} + 5.1 \text{ k}}$$

$$\text{or, } R_3 \approx 1.54 \text{ k} \text{ [use standard } 1.6 \text{ k]}$$

Using Equ.(18-41),

$$R_{SC} = \frac{0.65}{I_{o(max)}} = \frac{0.65}{100 \text{ mA}}$$

$$\text{or, } R_3 \approx 6.5 \Omega \text{ [use standard } 6.8 \Omega]$$

Comments: In this way regulated power supply within 7 V,

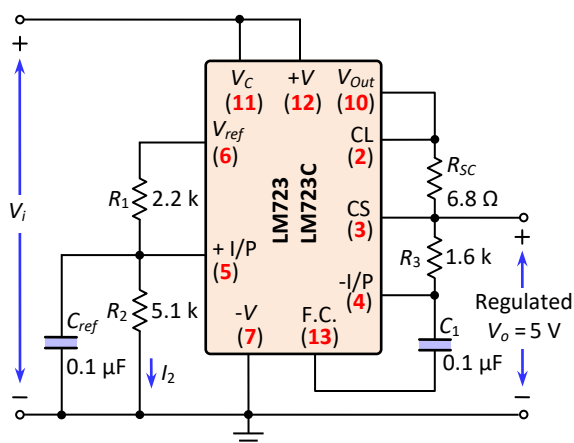


Fig.18-57: Regulated power supply designed in Example 18-11

18.15.4 High Current Positive Voltage Regulator

The above two regulators has the maximum output current capacity of only 150 mA. But using an external transistor, the output current capacity can be increased. The circuit arrangement is shown in Fig.18- (High voltage high current). The regulated output voltage of the IC is applied to the base of external transistor Q_3 . This resistor works as an emitter follower as used in discrete component voltage regulator [Fig.18-]. The overall output voltage will be same as given by Equ.(18-). The error amplifier will produce V_{BE3} Volt higher than V_o . The output voltage range of this circuit will be 7 V to 37 V regulated with output current up to 5 A.

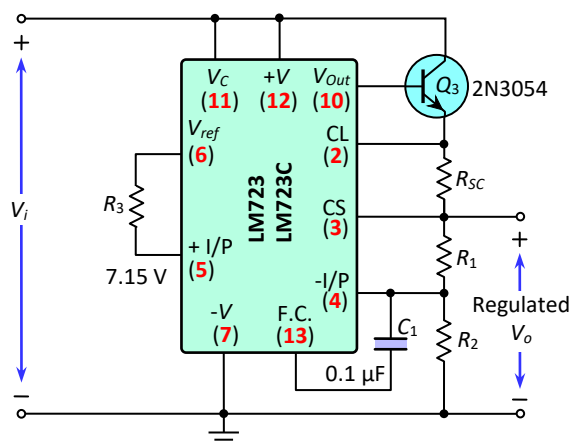


Fig.18-58: High current regulated power supply using LM723

18.15.5 Negative Voltage Regulator Using LM723

LM723 IC can also be used for negative voltage regulation. The circuit connection is given in Fig.18-

59. Here, $-V_i$ is the unregulated negative input voltage and V_o is the regulated negative output voltage. The value of $|V_i|$ should be always greater than $|V_o|$. Transistor Q_3 is the series-pass transistor which is driven by V_Z terminal (pin-9) of LM723. To understand the operation process, the output section of this circuit is redrawn in Fig.18-60 showing the internal components of LM723. From this figure, we find that the voltage to the inverting terminal of the error amplifier will be $-V_{ref}/2$, and that of the non-inverting terminal will be $-V_o \times R_2/(R_1 + R_2)$. But the error amplifier will adjust its output voltage $-V_{O(op)}$, until these two voltages are equal. When these two voltages are equal, that is, when

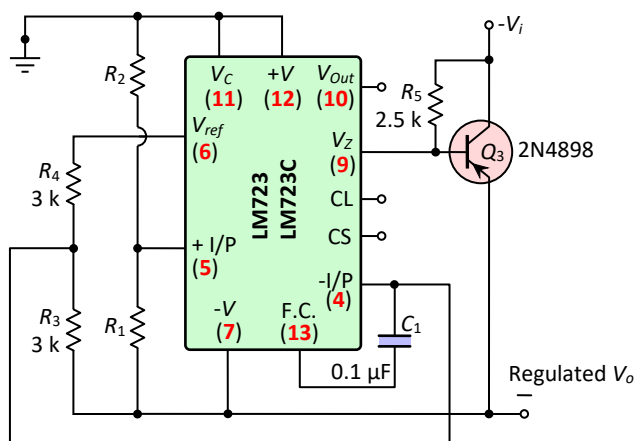


Fig.18-59: High current regulated negative power supply using LM723

$$\frac{-V_{ref}}{2} = -V_o \times \frac{R_2}{R_1 + R_2}$$

or,

$$-V_o = \frac{-V_{ref}}{2} \times \frac{R_1 + R_2}{R_2} \quad (18-44)$$

To produce this output voltage ($-V_o$), the output voltage of the error amplifier ($-V_{o(op)}$) should be a little bit higher than $-V_o$. To calculate this value, let us apply KVL in Fig.18- as shown by the dashed line. Both the error amplifier's output voltage ($-V_{o(op)}$), and the regulated output ($-V_o$) are measured with respect to ground. So starting from ground potential, we can write

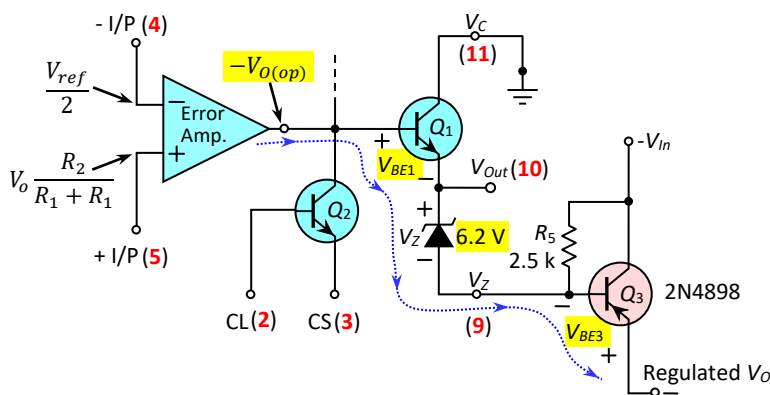


Fig.18-60: Output stage of the negative voltage regulator of Fig.18-47 with internal components of LM723 IC

$$-V_{o(op)} - V_{BE1} - V_Z + V_{BE3} + V_o = 0$$

If V_{BE1} and V_{BE3} are equal in magnitude,

$$-V_{o(op)} = -V_o + V_Z$$

For example, if the output voltage is $-V_o = -15$ V, the value of error amplifier's output will be,

$$-V_{o(op)} = -15 \text{ V} + 6.2 \text{ V} = -8.8 \text{ V}$$

Switch-Mode Power Supply

In series and shunt regulators, we studied in Sections, the control element, i.e. the transistor, operates in the active region and, therefore, a huge amount of power is lost in this transistor which results in low efficiency of these regulators. To overcome this problem Switch Mode Power Supply is used. Switch Mode Power Supply (SMPS) is also called Switching Mode Power Supply. As the name suggests, SMPS is a circuit which provides DC power to any electrical load and involves some kind of switching action. As we saw previously, linear power supplies become very bulky with increase in current ratings. So, we need something which will provide large amount of currents without taking a lot of space. SMPS is the solution for that.

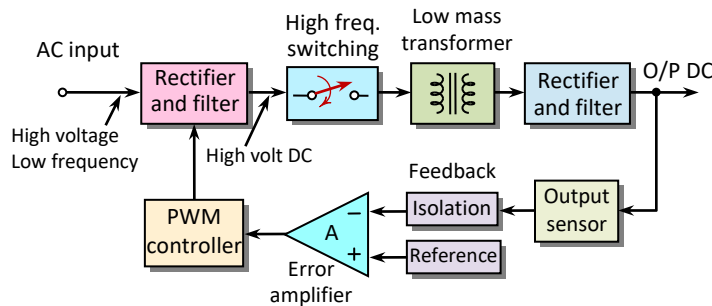


Fig.18-61: Block diagram of a switching type voltage regulator

SMPS works on a very high frequencies as compared to linear power supplies and as the size of transformer reduces with increase in frequency, the overall size of an SMPS become very small as compared to linear power supplies.

Basic blocks of an SMPS

There are basically five blocks in an SMPS :

- Input rectifier and filter
- Chopper
- Transformer
- Output rectifier and filter
- Feedback circuit

Working Principle of SMPS

SMPS works on high frequencies. We need to increase the frequency of 50 Hz input. So, convert the AC input to DC first and then chop it at high frequency to get the pulsating DC output which is then applied to rectifiers and filters. Feedback helps to maintain the level of output signal.

Advantages:

Small size

Low noise

High efficiency (80% to 95%)

Disadvantages:

High complexity

Produces EMI (Electromagnetic Interference)

Switch-mode power conversion topologies

Popular un-isolated topologies:

- ▶ Buck (or step-down) converter
- ▶ Boost (or step-up) converter
- ▶ Buck/Boost (step-up/down) converter

Popular isolated topologies

- ▶ Flyback converter
- ▶ Resonant converter

<https://www.youtube.com/watch?v=nmUug3WNIZI&t=1507s>

Switch-mode power conversion topologies

Depending on their advantageous features, a number of topologies are used in switching mode power supplies. The most common topologies are as follows.

Un-isolated topologies

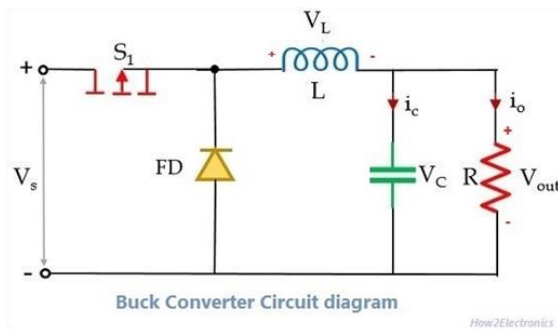
Un-isolated SMPS do not provide isolation between the input AC and the output DC section. So it is not safe to touch the output of this converter. The popular topologies used for un-isolated SMPS are given below.

- Buck (or step-down) converter
- Boost (or step-up) converter
- Buck-boost (step-up/down) converter

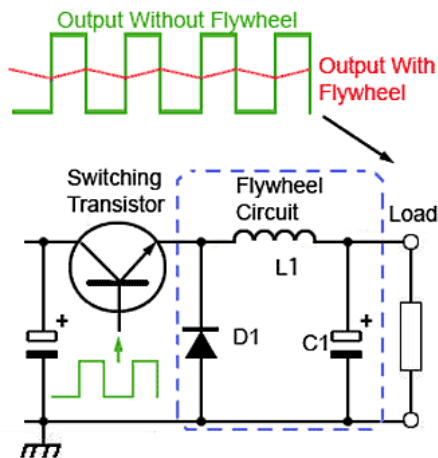
Isolated topologies

Isolated SMPS provides isolation between the input AC and the output DC section. So it is safe to touch the output of this converter. The popular topologies used for un-isolated SMPS are given below.

- Flyback converter
- Resonant converter



Step-down regulators are in fact, current



Step-Down Regulator

Step-down regulators are in fact, current boosters generally employed to drive current receiving loads such as permanent magnet, dc motor loaded with a constant -torque mechanical load. In this regulator, the output voltage is always less than the input voltage. This means that this type of regulator can be used as a dc step down transformer with highest efficiency.

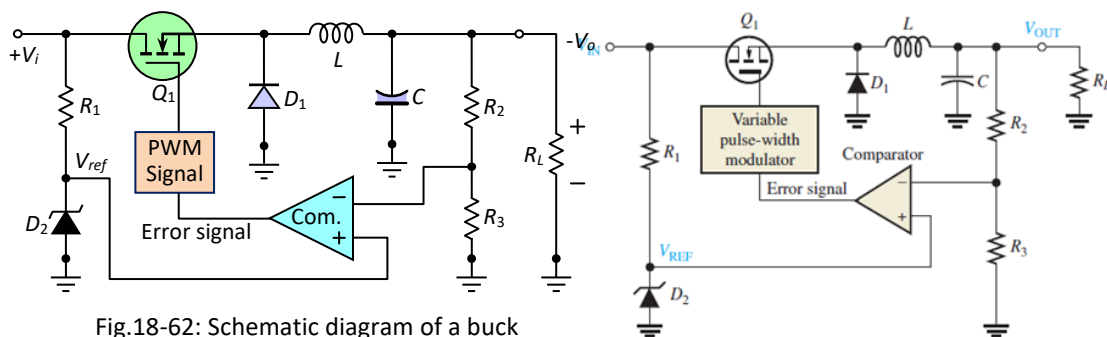


Fig.18-62: Schematic diagram of a buck switching regulator

► FIGURE 17-17

A basic step-down switching regulator.

Fig.20.26 (a) represents a simple step-down regulator power stage. The, output voltage is compared with a stable reference voltage anti the amplified error signal is used to generate a pulse width modulated waveform, which controls the on-off period of the switching transistor. When the switch is turned ON, current flows through the inductor and into output capacitor and the load. When the output voltage exceeds the reference voltage, the switching transistor is turned OFF. At this instant the stored energy in the inductor reverses its polarity, takes the path through the diode and sends the current into the load while the voltage is maintained by the capacitor. When all the stored energy in the inductor is used up, the capacitor discharges and the output voltage decreases. At this step, the switch is turned on and the process continues such that the output voltage is maintained very close to the reference voltage.

Theory

Let us consider the simplified Fig.20.26 (b) of a step-down converter to study the theory of operation. Let the switch S be kept oscillating with a period

or,	$T = T_{on} + T_{off}$	(18-45)
-----	------------------------	---------

When the switch, S , is ON, the diode is reverse biased and a voltage $V_L = V_i - V_o$ will be applied across the inductor, L . Then we can write,

or,	$V_L = V_i - V_o = L \frac{di_L}{dt}$	(18-46)
-----	---------------------------------------	---------

During this ON time, T_{on} , if the inductor current increases linearly from I_1 to I_2 , then we can write,

or,	$V_L = V_i - V_o = L \frac{(I_2 - I_1)}{T_{on}} = L \frac{\Delta i_L}{T_{on}}$	
or,	$\Delta i_L = \frac{V_i - V_o}{L} T_{on}$	(18-47)

When the switch is OFF, the inductor release its stored energy to the load through the diode, D_1 . Then,

or,	$-V_L = -V_o = -L \frac{di_L}{dt}$	(18-48)
-----	------------------------------------	---------

If the inductor current decreases from I_2 to I_1 in the time duration, T_{off} . Then,

or,	$V_o = L \frac{\Delta i_L}{T_{off}}$	
or,	$\Delta i_L = \frac{V_o}{L} T_{off}$	(18-49)

Combining Equ.(18-47) and Equ.(18-49), we get,

or,	$\frac{V_i - V_o}{L} T_{on} = \frac{V_o}{L} T_{off}$	
or,	$(V_i - V_o) T_{on} = V_o T_{off}$	
or,	$V_i T_{on} - V_o T_{on} = V_o T_{off}$	
or,	$V_o (T_{on} + T_{off}) = V_i T_{on}$	
or,	$V_o = \frac{T_{on}}{T_{on} + T_{off}} V_i$	(18-50)
Using Equ.(18-45),	$V_o = \frac{T_{on}}{T} V_i$	(18-51)

But T_{on}/T is the duty cycle, D , of the switching element. Therefore,

or,	$V_o = DV_i$	(18-52)
-----	--------------	---------

As $1/T$ is the frequency (f) of switching element, Equ.(18-) can also be written as,

or,	$V_o = f T_{on} V_i$	(18-53)
-----	----------------------	---------

From Equ.(18-52), we find that V_o is a linear function of the input voltage. As the value of the duty cycle (D) is always less than unity, we can find that the output voltage, V_o , is always less than the input voltage, V_i . For this reason, this converter is called step down DC-to-DC converter. Capacitors C_1 and C_2 are used to absorb input and output voltage variations and to provide current boosting and smoothing.

Step Up Converter

Step up regulators are voltage booster regulators. In this regulator, the output voltage is always higher than the input voltage. Fig shows a step up regulator. The conduction of the transistor Q_1 is controlled by the pulse width modulated signal. When the transistor is turned ON, the current flows through the inductor and the energy is stored in it. When the switch is turned OFF, the stored energy in the inductor tends to release and its polarity changes such that it adds to the input voltage. Thus the voltage across the inductor and the input voltage are in series and together charge the output capacitor to a voltage higher than the input voltage, V_i .

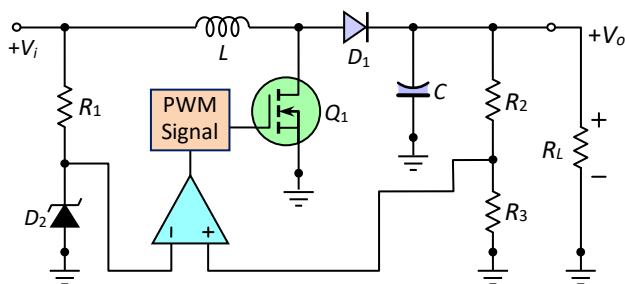
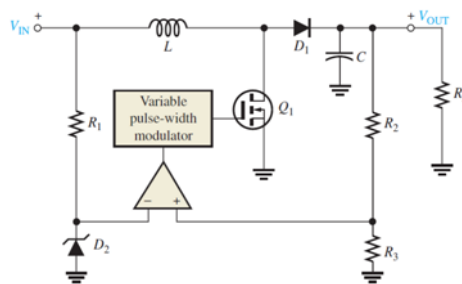


Fig.18-63: Schematic diagram of a boost switching regulator



► FIGURE 17-20

Basic step-up switching regulator.

Theory

Let us consider the simplified diagram of Fig.18-. Let us assume that the switching time period of the transistor is

or,	$T = T_{on} + T_{off}$	(18-54)
-----	------------------------	---------

When the switch, S , is ON, a voltage $V_L = V_i$ is applied across the inductor, then,

or,	$V_L = V_i = L \frac{di_L}{dt}$	(18-55)
-----	---------------------------------	---------

If the inductor current rises linearly from I_1 to I_2 during the ON time, T_{on} , then,

$$V_L = V_i - V_o = L \frac{(I_2 - I_1)}{T_{on}} = L \frac{\Delta i_L}{T_{on}}$$

or,	$\Delta i_L = \frac{V_i}{L} T_{on}$	(18-56)
-----	-------------------------------------	---------

When the switch is OFF, the inductor releases its stored energy $V_L = V_o - V_i$ through the diode, D_1 , then

	$V_L = V_o - V_i = L \frac{di_L}{dt}$	
--	---------------------------------------	--

If the inductor current falls linearly from I_1 to I_2 during the OFF time of the switch, T_{off} , then,

	$V_L = V_i - V_o = L \frac{(I_1 - I_2)}{T_{off}} = L \frac{\Delta i_L}{T_{off}}$	
or,	$\Delta i_L = \frac{V_o - V_i}{L} T_{off}$	(18-57)

Equating Equ.(18-) and Equ.(18-), we get,

or,	$\frac{V_o - V_i}{L} T_{off} = \frac{V_i}{L} T_{on}$	
or,	$V_o T_{off} - V_i T_{off} = V_i T_{on}$	
or,	$V_o T_{off} = V_i (T_{on} + T_{off})$	
or,	$V_o = \frac{(T_{on} + T_{off})}{T_{off}} V_i$	
Using Equ.(18-54),	$V_o = \frac{T}{T_{off}} V_i$	
or,	$V_o = \frac{1}{T_{off}/T} V_i = \frac{1}{(T - T_{on})/T} V_i$	
$\therefore \frac{T_{on}}{T} = D$	$V_o = \frac{1}{1 - D} V_i$	(18-58)

From Equ.(18-), we find that the output voltage, V_o will be greater than the input voltage. For this reason this converter is called step up or boost converter. Here also the output is a linear function of input. The function of the capacitors, C_1 and C_2 is the same as described for buck converter.

Buck/Boost converter

A polarity inverting regulator provides an output voltage of opposite polarity to that of the input voltage. This regulator can also produce output voltage which may be less than or greater than the input voltage and hence this regulator is also called buck-boost regulator. Fig.18- shows a polarity inverting regulator. The conduction of the transistor is controlled by the pulse width modulated signal. When the transistor is turned ON, the current flows through the transistor collector and the inductor L and energy is stored in the inductor. When the switch is turned OFF, the stored energy in the inductor tends to collapse and its polarity changes such that the diode conducts resulting in a transfer of inverted voltage to the output circuit.

Theory

Let us consider the simplified circuit diagram of the buck-boost converter shown in Fig.18- and the switch oscillates with a time period of $T = T_{on} + T_{off}$. When the switch is ON, a voltage $V_L = V_i$ is applied across the inductor, then

or,	$V_L = V_i = L \frac{di_L}{dt}$	(18-59)
-----	---------------------------------	---------

If the inductor current rises linearly from I_1 to I_2 during the ON time, T_{on} , then,

$$V_i = L \frac{(I_2 - I_1)}{T_{on}} = L \frac{\Delta i_L}{T_{on}}$$

or,	$\Delta i_L = \frac{V_i}{L} T_{on}$	(18-60)
-----	-------------------------------------	---------

When the switch is OFF, the inductor releases its stored energy $V_L = V_o$ will be transferred to the output, then

	$V_L = V_o = -L \frac{di_L}{dt}$	
--	----------------------------------	--

If the inductor current falls linearly from I_2 to I_1 during the OFF time of the switch, T_{off} , then,

$$V_o = -L \frac{(I_1 - I_2)}{T_{off}} = -L \frac{di_L}{T_{off}}$$

or,	$\Delta i_L = -\frac{V_o}{L} T_{off}$	(18-61)
-----	---------------------------------------	---------

Equating Equ.(18-) and Equ.(18-), we get,

or,	$\frac{V_i}{L} T_{on} = -\frac{V_o}{L} T_{off}$	
or,	$V_i T_{on} = -V_o T_{off}$	
or,	$V_o = -\frac{T_{on}}{T_{off}} V_i$	
or,	$V_o = \frac{T_{on}}{(T - T_{on})} V_i$	
or,	$V_o = \frac{T_{on}/T}{(T - T_{on})/T} V_i$	
$\therefore \frac{T_{on}}{T} = D$	$V_o = \frac{D}{(1 - D)} V_i$	(18-62)

From Equ.(18-) we find that the output voltage is linear function of input voltage. Moreover, V_o can be greater than or less than the input voltage. If the value of the duty cycle (D) is greater than 0.5, then $V_o > V_i$. But If the value of the duty cycle (D) is less than 0.5, then $V_o < V_i$. That is we get bucking and boosting properly from this converter. Therefore, this converter is called a buck-boost converter. The negative sign in Equ.(18-) indicates that the voltage will be of opposite polarity of the input. For this reason this circuit is also called **polarity inverter**.

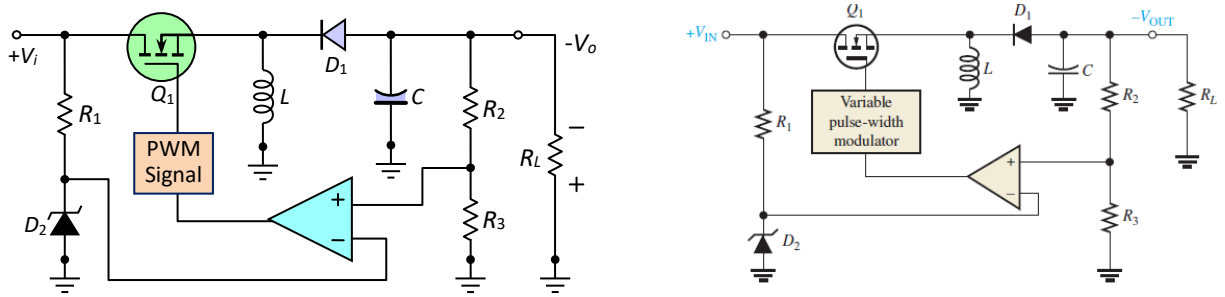
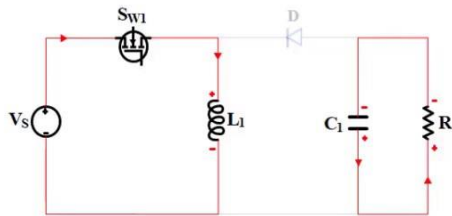
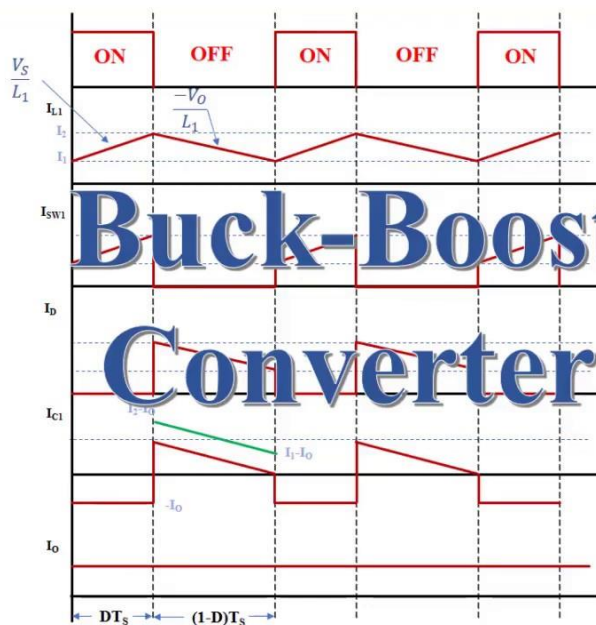
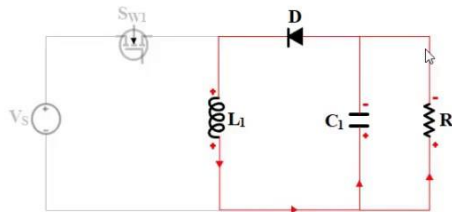


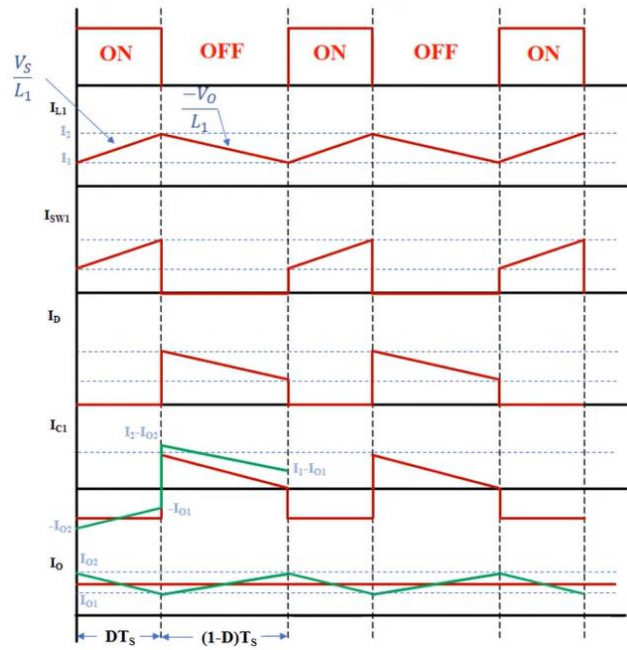
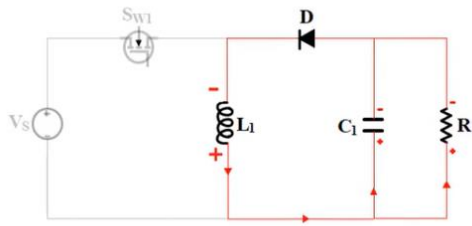
Fig.18-64: Schematic diagram of a buck-boost switching regulator

Mode I



Mode II



Mode II

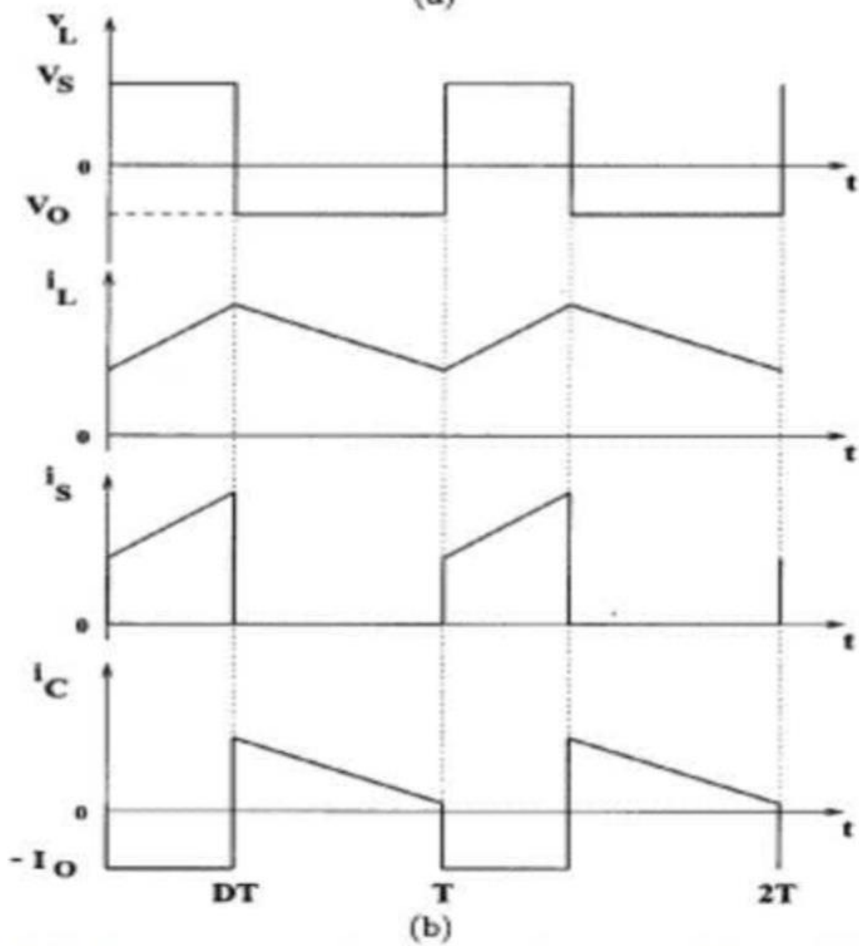
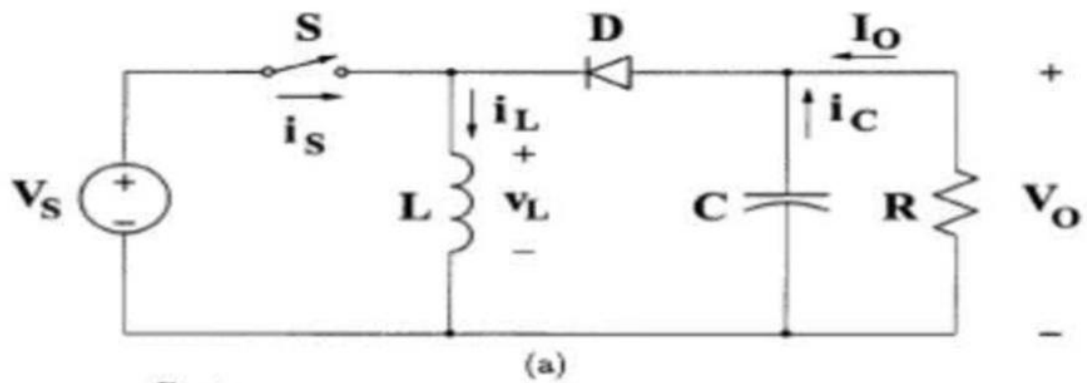
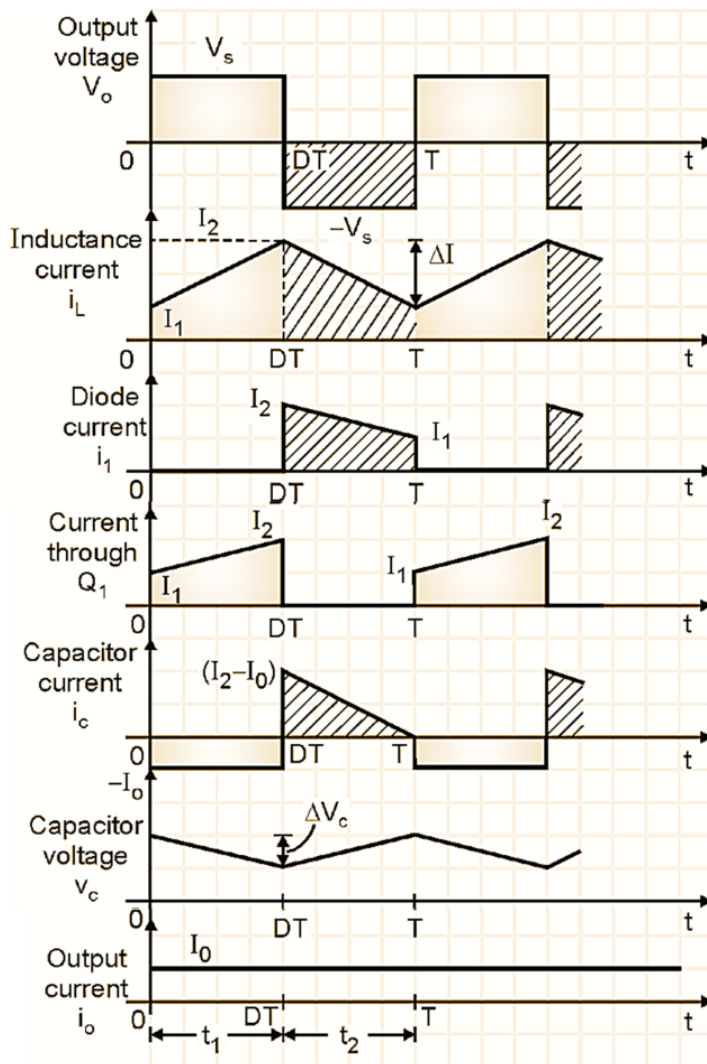


Fig.1 Buck-boost converter: (a) circuit diagram; (b) wave-



18.16 Flyback Converter

The name flyback comes from the analog TV circuit, in which there was a beam of electrons that sweeps on the screen and produces the picture. A high voltage converter circuit was used to accelerate the electron beam and sweep it on the screen. As the beam *flies back* to its starting position rapidly, the converter circuit is called *flyback converter*.

The flyback converter belongs to the *primary switched converter* family. It uses a coupled inductor, which provides isolation between input and output. The main difference of coupled inductor from transformer is that when current flows through the primary coil of coupled inductor it stores energy but in transformer it does not (in this text it is also referred to as transformer).

Flyback converters are used in nearly all mains supplied electronic equipment for low power consumption, up to approximately 300W. Examples of which are televisions, personal computers,

printers, etc. Flyback converters have a remarkably low number of components compared to other SMPSs, they also have the advantage that several isolated output voltages can be regulated by one control circuit (see Fig.18-).

Fig.18- shows the basic circuit of a flyback converter. The transistor (MOSFET) Q_1 is turned ON and OFF by the pulse-width-modulated control voltage V_{ct} . During the ON-time of the transistor, the voltage of the primary inductor v_p is equal to the input voltage, V_i which results in the primary current, i_p increasing linearly. During this ON-state, energy is stored in the primary coil. As shown in Fig.18-, the polarity of the secondary voltage (v_s) is such that the diode (D_1) is reverse biased. So the secondary current is zero. During this time the capacitor, C_2 , discharges and provides energy to the load, R_L .

When the transistor is turned OFF, the primary coil releases its energy and its voltage

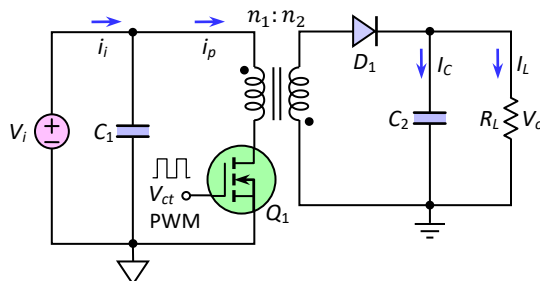


Fig.18-65: Schematic diagram of flyback

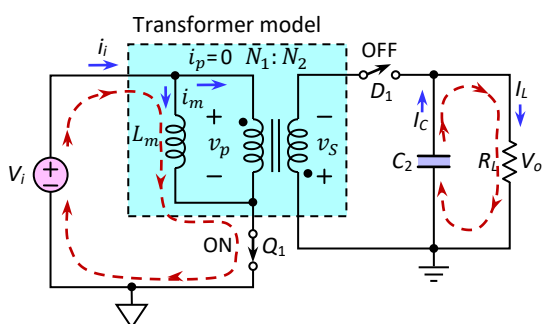


Fig.18-66: Flyback converter during ON state

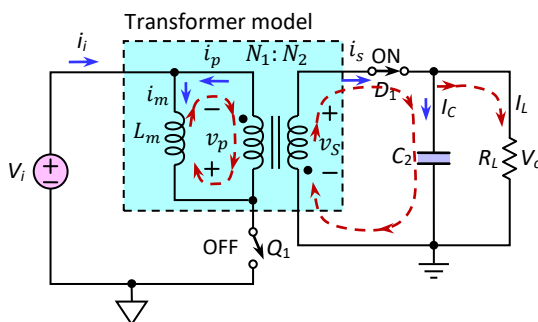


Fig.18-67: Flyback converter during OFF state

<https://www.youtube.com/watch?v=Oqn9WMPqAZU&t=1611s>

($v_p = L di/dt$) polarity is reversed. Current I_1 continues to flow through the magnetising inductor (L_m). At the time the voltage polarity of the secondary coil is also reversed and the diode becomes forward biased. So, the secondary current I_2 flows through the capacitor and the load. During the ON state, the capacitor provides current to the load.

During the ON-state of the transistor the drain-source voltage V_{DS} is equal to zero. During the OFF-state of the transistor, the output voltage V_o will be transformed back to the primary side and the drain-source voltage theoretically steps up to $V_{DS} = V_i + V_o \times N_1/N_2$. If a voltage of 230 V/50 Hz is used V_{DS} will jump up to approximately 700 V. In practice this voltage will be even higher due to the self

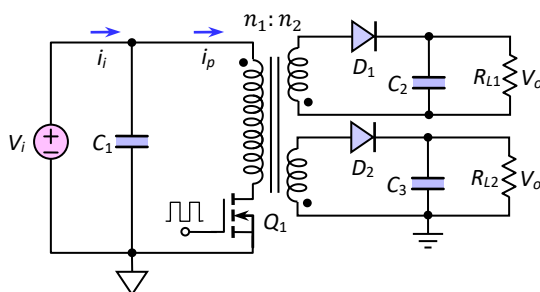


Fig.18-68: Flyback converter for multiple outputs

induction of the leakage inductance. To allow for this effect the minimum rated drain-source breakdown voltage of the transistor must be 800 V.

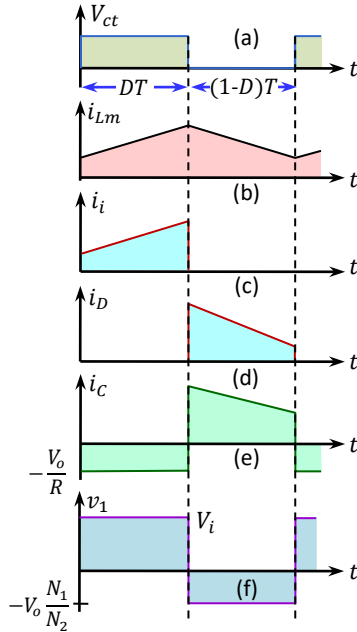


Fig.18-69: Waveforms of flyback converter

From Fig.18-(f),

	$V_i D T - V_o \left(\frac{N_1}{N_2} \right) (1 - D) T = 0$	
or,	$V_i D - V_o \left(\frac{N_1}{N_2} \right) (1 - D) = 0$	
or,	$V_o \left(\frac{N_1}{N_2} \right) (1 - D) = V_i D$	
or,	$\boxed{\frac{V_o}{V_i} = \frac{D}{(1 - D)} \left(\frac{N_2}{N_1} \right)}$	(18-63)

Example 18-12

A flyback converter like that of Fig.18- has the following parameters: $V_i = 24$ V, $N_1/N_2 = 3:1$, $L_m = 0.5$ mH, $R_L = 5$ Ω , $C_2 = 200$ μ F, $f = 40$ kHz, and $V_o = 5$ V. Determine i) the required duty cycle (D), ii) the minimum, maximum and average current through L_m , and iii) the output ripple voltage. Assume all the components are ideal.

Solution:

From the output voltage and load resistance we find the output current $I_o = V_o/R_L = 5\text{ V}/5\ \Omega = 1\text{ A}$. The power delivered to the load will be,

$$P_o = \frac{V_o^2}{R} = V_o I_o = 5\text{ V} \times 1\text{ A} = 5\text{ Watt}$$

i) Required duty cycle:

Using Equ.(18-),	$\frac{V_o}{V_i} = \frac{D}{(1-D)} \left(\frac{N_1}{N_2} \right) = \frac{5}{24}$	(18-64)
	$\frac{5}{24} = \frac{D}{(1-D)} \left(\frac{1}{3} \right)$	
	$D \approx 0.385$ [Ans.]	

ii) Currents in L_m

For the ideal situation, the power supplied to the primary of the transformer will be equal to the output power which is 5 W. So the average current to the primary will be, 5/24 A.

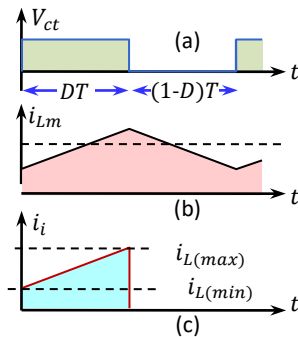


Fig.18-70: Waveforms of flyback converter

	$i_{s(avg)} = \frac{i_{L(min)} + i_{L(max)}}{2} \times \frac{DT}{T} = \frac{5}{24}$	
	$\frac{i_{L(min)} + i_{L(max)}}{2} \times D = \frac{5}{24}$	
	$i_{L(avg)} = \frac{i_{L(min)} + i_{L(max)}}{2} = \frac{5}{24} \times \frac{1}{D}$	
	$i_{L(avg)} = \frac{i_{L(min)} + i_{L(max)}}{2} = \frac{5}{24} \times \frac{1}{0.385}$ $\approx 540\text{ mA}$ [Ans.]	

	$\Delta i_L = \frac{V_i DT}{L_m} = \frac{24 \times 0.385}{500\ \mu\text{H}} \times \frac{1}{40\text{ kHz}} \approx 460\text{ mA}$	
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	$i_{L(min)} = i_{L(avg)} - \frac{\Delta i_L}{2} = 540 \text{ mA} - \frac{460 \text{ mA}}{2} = 310 \text{ mA [Ans.]}$	
	$i_{L(max)} = i_{L(avg)} + \frac{\Delta i_L}{2} = 540 \text{ mA} + \frac{460 \text{ mA}}{2} = 770 \text{ mA [Ans.]}$	

iii) Output Ripple Voltage

	$\Delta V_o = \frac{I_o DT}{C} = \frac{1 \text{ A} \times 0.385}{200 \mu\text{F} \times 40 \text{ kHz}} \approx 0.048$	
	$\text{Output ripple (\%)} = \frac{\Delta V_o}{V_o} \times 100 = \frac{0.048 \text{ V}}{5 \text{ V}} \times 100 \approx 1\% \text{ [Ans.]}$	

<https://www.youtube.com/watch?v=wYS25HUaEqI>

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Dentotal

Khaja Md. Shahidullah

Rasonance Converter

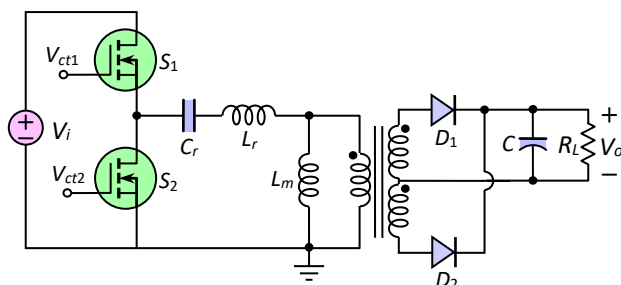


Fig.18-71: Schematic diagram of a Resonant switching regulator

The circuit diagram of a resonant converter is shown in Fig.18-. It consists of mainly three stages. The first stage consists of two MOSFETs and is called the switching section. It generates a pulsating voltage or current from a DC input voltage with a specific frequency. This waveform will be in the form of a square wave. The next section is the series resonant circuit consisting of a capacitor (C_r) and an inductor (L_r). It modulates the gain amplitude by changing the pulsating voltage. It also smooths out the square wave power signal which is being fed to the rectifier and it is very helpful for soft switching of the switching elements. Soft switching helps to reduce the switching power

loss. The resonant circuit only select the first harmonic or the fundamental frequency, and thus, perform the task of smoothing. This converter can work in the range of MHz. This signal is further step up/down by a centre-tape transformer. The last stage is the full-wave rectifier with a capacitor filter. This stage converts the sinusoidal power into pulsating DC and the capacitor filter produces a DC output voltage.

The resonant converters provide high efficiency at high power also the overall size of the converter is smaller than the buck-boost converter. They are widely used in servers, telecoms & high-power consumer electronics.

18.17 SMPS Using DK1203 IC

The DK1203 is a secondary side flyback type AC-DC Switch Mode Power Controlling IC. With integrated 700 V power transistor, self-power supply circuit and integrated CMOS circuit design, lots of external components are saved, transformer design is simple, only two windings are needed for isolating the output circuit.

18.17.1 MAIN FEATURES

- Wide range (85 V— 264 V) AC power input
- Integrated high voltage (700 V) power transistor
- Integrated constant high voltage current driving circuit
- Self-power supply circuit design, no need for additional electricity supply
- 65 KHz PWM switching frequency
- Low standby power loss ($< 0.3\text{ W}$)
- Built in frequency jittering function
- Internal ramp compensation circuit
- Internal PMW oscillation circuit
- Over current, over temperature, over voltage, output short circuit and secondary side open circuit protection.

18.17.2 Applications

- AC/DC power adapters
- DVD/VCD power supply
- Air conditioner power supply
- Electromagnetic oven power supply
- DVB power supply
- LED driver applications
- TV/Monitor power supply

18.17.3 Pin Configuration

DK1203 ICs are fabricated with different packages. Fig.18-, shows the pin-configuration of DIP-8 package. Though the IC has 8 pins, two pins (pin no. 4 and 6) are not used (NC). Pin 2 is V_{CC} and pin 3 is the GND connection. The functions of these pins are described in the Table 18-3.

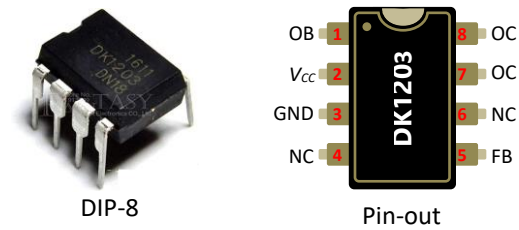
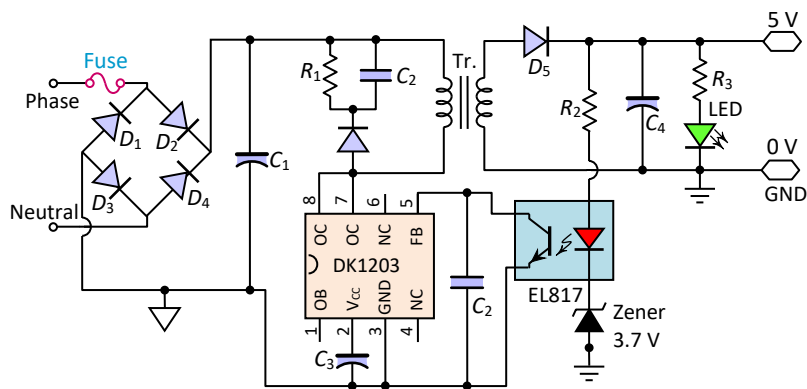


Fig.18-72: Pin configuration of LM723 IC

Table 18- [Courtesy of DONGKE SEMICONDUCTOR]

Pin No.	Pin Name	Function
1	OB	Startup pin. Internal startup circuit to contact with OC pin
2	V _{CC}	Power supply of control circuits, contacted with an external grounded capacitor of 10 μ F~100 μ F
3	GND	Ground reference
4	NC	Empty pin without internal connection
5	FB	Feedback control pin, connected with 1 nF ~10 nF grounded capacitor, coupler grounded to control output
6	NC	Empty pin without internal connection
7	OC	Output pin, connected with internal high voltage Collector point and switch mode transformer
8	OC	Same as pin 7



Approved in the C&D committee meeting

Floyd Chapter 17 page

Malvino Chapter 22

7805 internal diagram

<http://www.righto.com/2014/09/reverse-engineering-counterfeit-7805.html>